Localized Matched-Subspace Detection for Functional Ultrasound and Ultrafast Doppler Imaging

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Abstract

Functional ultrasound and ultrafast Doppler maps can remain vulnerable to nuisance-driven false positives after conventional clutter suppression, especially in the low-false-positive regime where rare artifacts can dominate interpretation. We study a post-residual detector-head framework for beamformed fUS slow-time data. The fixed default is a band-limited matched-subspace score; an adaptive extension turns on tile-local whitening only where residual guard-band energy indicates localized clutter; and a fully whitened specialist is kept for regimes in which full local covariance adaptation earns its extra complexity. Quantitative claims are limited to labeled brain simulations, a structurally labeled flow phantom, a held-out SIMUS/PyMUST moving-scatterer benchmark, and real-IQ audits with explicitly bounded scope. Across the labeled brain stress tests and Gammex phantom, the matched-subspace family retains nonzero sensitivity at strict low-FPR operating points where PD/Kasai comparators can collapse. Simple same-residual ablations show that the main labeled-brain gain is not explained by whitening alone: total whitened power without Doppler band partition still collapses in the strict tail, whereas the band-limited matched-subspace score remains effective. On the held-out SIMUS/PyMUST structural benchmark, the fixed non-whitened matched-subspace score is the strongest fixed downstream head among the evaluated comparators, while held-out stress rows and fixed-threshold transfer checks show where the adaptive and fully whitened variants help beyond that default. On real in-vivo rat-brain IQ, a localization-derived PALA vascular reference yields a modest but consistent structural advantage for a fully whitened specialist over PD and Kasai on a same-acquisition structural audit. This real-IQ result is structural rather than task-activation evidence. Overall, these results support downstream detector-head design and strict-tail evaluation as practical complements to conventional clutter suppression.

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Evidence at a glance. The paper makes four kinds of claims, from controlled labeled detection to more limited real-IQ corroboration. The validation ladder below appears before the method section so readers can see the scope of the evidence before the technical details.

Immediate external test. The clearest next validation step is prospective labeled *in vivo* evaluation on raw high-PRF IQ with an independent anatomical or task anchor and a genuinely fixed threshold-transfer workflow across repeated sessions.

Regime	Independent anchor	What it establishes in this paper
Brain simulation stress tests	simulator truth	controlled mechanism checks and strict-tail sensitivity under labeled nuisance stress
Gammex phantom	B-mode structural labels	structural ROC on real phantom data with independent geometry
Held-out SIMUS/PyMUST benchmark	simulated label masks in use-case-motivated scenes	main held-out structural benchmark, stress-frontier confirmation, and fixed-calibration transfer
ULM structural audit	published localization-derived vascular reference	bounded same-acquisition structural audit on real in-vivo IQ
ULM motion + Shin audits	no independent task labels	bounded robustness and corroborative real-IQ evidence only

Table 1: Compact validation ladder for the paper. The evidence progresses from labeled simulation and structurally labeled phantom data to a held-out use-case-motivated synthetic benchmark and one same-acquisition structural audit on open real in-vivo IQ. The fuller claim-support map appears later in Table 4.

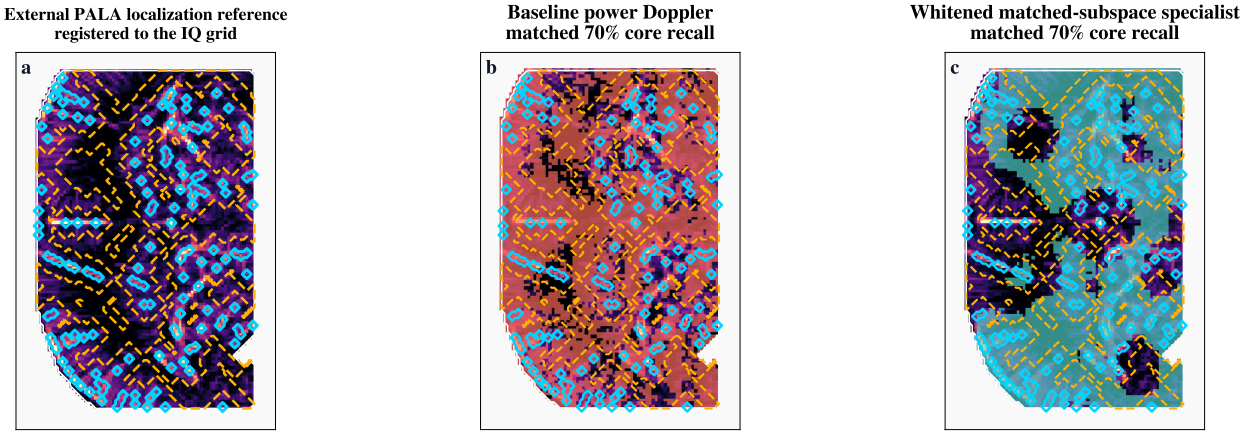

Figure 1: Localization-derived structural-label result on ULM Zenodo 7883227 using the published PALA rat-brain localization rendering as an external vascular reference. Left: the PALA-derived vessel-core and perivascular-shell masks after registration back to the IQ grid. Middle and right: the same audited 64-frame MC–SVD residual window (block 001, frames 128:192), thresholded at matched 70% vessel-core recall for baseline power Doppler and the prespecified no-registration whitened matched-subspace specialist. This figure is intentionally visual; the formal held-out quantitative comparison is reported in Table 2 immediately below.

Score	AUC (H1 vs H0)	TPR@$\alpha = 10^{-1}$	TPR@$\alpha = 10^{-2}$	TPR@$\alpha = 10^{-3}$
Baseline (power Doppler)	0.503 [0.493,0.513]	0.105 [0.097,0.113]	0.009 [0.007,0.011]	0.000 [0.000,0.000]
Baseline (log-power Doppler)	0.503 [0.493,0.513]	0.105 [0.097,0.113]	0.009 [0.007,0.011]	0.000 [0.000,0.000]
Baseline (Kasai lag-1 magnitude)	0.502 [0.492,0.512]	0.106 [0.098,0.114]	0.008 [0.007,0.011]	0.000 [0.000,0.000]
Matched-subspace detector (whitened specialist)	0.526 [0.514,0.536]	0.147 [0.141,0.152]	0.023 [0.018,0.027]	0.003 [0.001,0.007]

Table 2: Structural-label ROC on real in vivo rat-brain IQ from ULM Zenodo 7883227. For each evaluation block, vessel and background masks are derived once from an external super-resolved vascular reference derived by registering the published PALA rat-brain ULM example back onto the Zenodo IQ grid and downsampling the registered MatOut rendering to the short-window evaluation lattice; short 64-frame windows from the held-out block are then scored without additional frame-to-frame registration on the same MC–SVD residual cube using different downstream detector heads. Entries report window-level means with 95% bootstrap CIs over windows. Reproducibility details are provided in Supplementary Material.

1 Introduction

Functional ultrasound (fUS) is now used for vascular and functional mapping in settings ranging from preclinical neuroscience to awake surgery and cranial-window human imaging. In these settings, the practical problem is often not average sensitivity alone. A small number of nuisance-driven false detections can distort vascular readouts, contaminate functional maps, and undermine confidence in interpretation. Tissue motion, phase instability, and aliased high-velocity flow therefore matter most when they survive ordinary clutter suppression and populate the extreme right tail of the detector output.

We address that problem with a localized matched-subspace detector framework for post-residual fUS. In plain terms, the method asks a local question after ordinary clutter suppression: within a small neighborhood, is the remaining slow-time signal concentrated in a prespecified flow band, or is it more consistent with structured nuisance? Among the downstream heads tested on the held-out SIMUS/PyMUST structural benchmark, a band-limited matched-subspace score without local whitening is the best fixed choice. A second detector keeps that score by default and turns on local whitening only when residual guard-band structure indicates localized clutter. A third, fully whitened detector is kept for regimes where full covariance adaptation is worth the extra complexity. In other words, clutter suppression is only the first stage. A coarse residualizer removes the dominant clutter, and a second scoring stage decides whether the remaining signal looks more like microvascular flow or structured nuisance. Figure 1 previews the clearest real-data version of that result: on real in-vivo rat-brain IQ, the fully whitened specialist is more concentrated on the localization-derived vascular core and less contaminated by the adjacent shell than conventional Doppler scoring. That is a different endpoint from the SIMUS nuisance-control benchmark, so it is not surprising that the winning detector is different there. We use matched-subspace as the plain-language term throughout the results. The radar lineage is only background context: this method performs localized post-beamformed slow-time scoring, not array-domain radar processing.

Terminology. Throughout the abstract, introduction, results, figures, and tables, we refer to the three detector modes as the fixed matched-subspace detector, the adaptive detector, and the fully whitened detector. The abbreviations STAP, CFAR, and KA are retained only in Methods and notation for historical context, where they refer to local slow-time whitening, empirical negative-tail

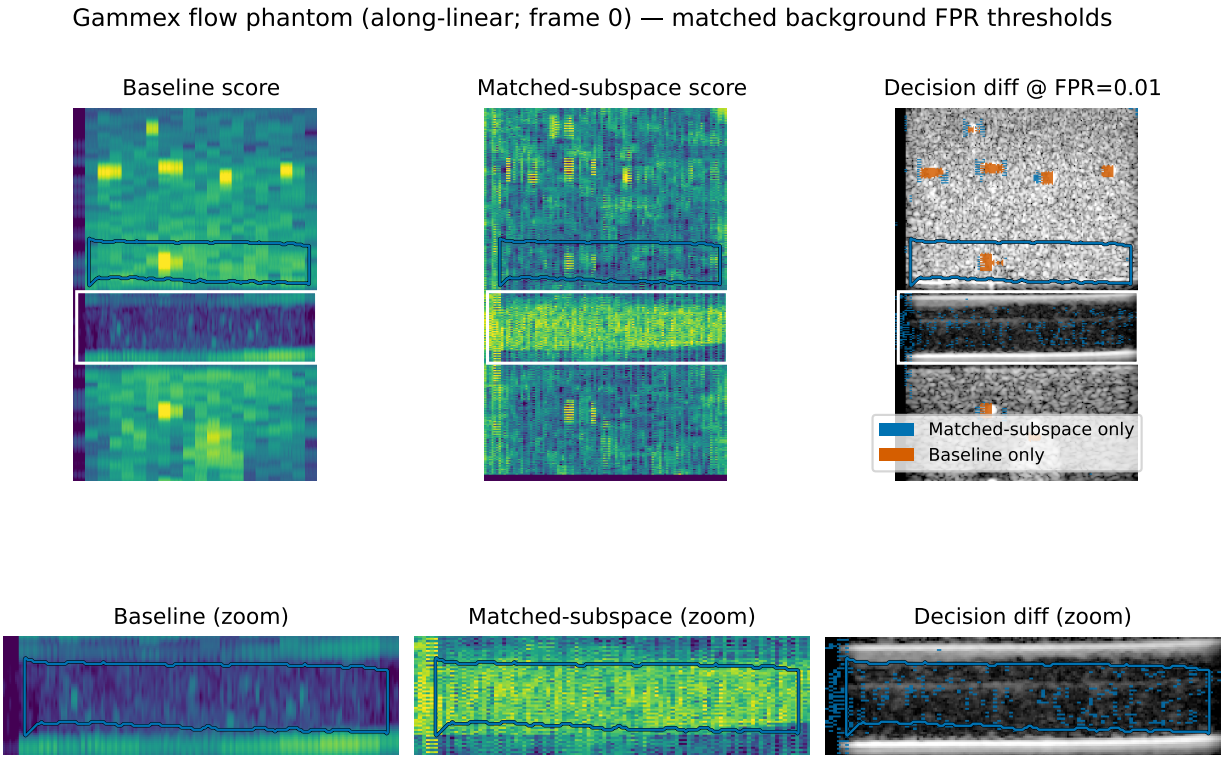

Figure 2: Leading structural-fidelity visual on a structurally labeled Gammex flow phantom acquisition from the open Doppler Twinkling Artifact database (labels derived from B-mode-only tube segmentation). Top row: baseline score map S_{base} (SVD band-pass clutter suppression followed by power Doppler), matched-subspace score map before the optional shrink-only penalty, and a matched-FPR decision-difference panel at a fixed background false-positive rate (10^{-2}; chosen here for visualization), showing pixels detected only by the matched-subspace detector versus only by the baseline. Bottom row: zoomed region of interest. The tube contour is overlaid for reference.

calibration, and an optional shrink-only regularizer, respectively.

This is not only an extreme-tail effect. On the representative Gammex frame in Figure 2, matching the background FPR at 10^{-2} yields zero lumen recovery for the baseline score, whereas the matched-subspace detector score recovers about 10% of lumen pixels at the same matched background allowance. We therefore use the stricter 10^{-3}–10^{-4} operating points mainly as a stress test that exposes nuisance sensitivity cleanly, not because the practical relevance begins only there.

We also study a conservative shrink-only regularization layer, but only as an optional add-on: it can compress extreme-tail artifacts without score inflation, and the main paper does not rely on it for its primary efficacy claims (details in the Supplementary Material). This problem matters now because fUS is no longer only a preclinical proof-of-concept modality. It is already used in intra-operative, neonatal, cranial-window, and emerging mobile settings [1, 8, 9, 20–22], where rare nuisance-driven false positives can matter more than average image quality.

The main claim of the paper is simple: after conventional clutter suppression, a localized matched-subspace downstream score can reduce nuisance leakage better than PD/Kasai-based scoring on the evaluated tasks. The rest of the paper asks where that simpler fixed score is already

sufficient, where an adaptive whitening switch is useful, and where a fully whitened specialist mode earns its extra complexity. The validation ladder in Table 1 is placed before the methods for exactly that reason: it gives readers the claim scope and evidence order up front, before the notation and benchmark detail.

Contributions.

- A post-residual localized matched-subspace detector family for beamformed fUS slow-time data, centered on a fixed band-limited head and extended by an adaptive whitening switch, a fully whitened specialist mode, and an optional shrink-only penalty layer.
- Held-out evidence that the fixed non-whitened matched-subspace score is the best-performing fixed head among the evaluated downstream heads on the SIMUS/PyMUST structural benchmark, together with harder stress-frontier results showing where the adaptive and fully whitened extensions add value.
- A cross-regime detector-head evaluation discipline that separates residualization from downstream scoring, uses strong ultrasound-native baseline families, includes simple same-residual detector ablations, and emphasizes strict-tail nuisance control rather than image appearance alone.
- Bounded deployment and translation evidence, including fixed-calibration threshold-transfer checks, reproducible latency measurements, and a localization-derived structural-label audit on real in-vivo rat-brain IQ.

Scope and claims. Claims are stratified by data regime and summarized first in the compact validation ladder of Table 1, then in the fuller claim-support map of Table 4. Quantitative detection efficacy is limited to regimes with independent H_1/H_0 labels or a structural reference that is external to the short-window detector outputs being scored. Open in-vivo IQ and PD-only datasets are otherwise used for robustness and translation audits. The ULM section adds one structural audit using a localization-derived vascular reference, not a task-activation benchmark.

Next external test. The practical collaboration target is now straightforward: prospective labeled *in vivo* validation on raw high-PRF IQ with an independent anatomical or task anchor and repeated sessions suitable for fixed-threshold transfer. Later in the paper we explain why that benchmark is still not available in one standardized public package, even though simultaneous fUS–Neuropixels, raw fUS plus two-photon, and early human task-fUS resources are beginning to make that next study realistic.

Canonical benchmark names. To keep the narrative compact, the remainder of the paper uses a small fixed set of benchmark names: *brain simulation stress tests*, *Gammex phantom*, *held-out SIMUS/PyMUST benchmark*, *ULM motion audit*, *ULM structural audit*, and *Shin real-IQ audit*. Development-generation labels, internal profile identifiers, and reproduction-script names are kept in the Supplementary Material and repository manifests rather than in the main reader path.

Organization. Section 2 defines the detection problem and score definitions. Section 3 presents the localized detector, with the fixed head first and the adaptive/fully whitened extensions second; the optional shrink-only regularizer is summarized briefly and left to the Supplementary Material for

full details. Section 4 describes the prespecified operating settings and integrity controls. Section 5 reports results, followed by discussion and conclusion.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Supplementary Material).

2.1 Related context

Ultrasound clutter suppression. Spatiotemporal SVD clutter filters are a standard tool in ultrafast Doppler and fUS [6, 7, 11, 12, 18]. Real-time feasibility has been studied via randomized/approximate decompositions and GPU implementations [16, 17]. Robust PCA and tensor/HOSVD variants have also been studied for clutter suppression and motion robustness [30, 31, 44]. We therefore benchmark against representative public ultrasound-native clutter-filter families, including adaptive-global SVD, local block-wise SVD, adaptive-local SVD, RPCA, HOSVD, and conventional Doppler score heads; our claim is superiority over those evaluated baseline families under the paper's low-FPR detector-head protocol, not blanket state-of-the-art superiority over every published pipeline.

What is new relative to Yu, Baranger, and Song. Eigen-based clutter filter design for ultrasound color flow imaging and block-wise/local adaptive clutter filters are closely related ultrasound-native approaches to covariance/eigenspace adaptation [13, 14, 16]. Song *et al.* demonstrate local clutter filtering in *in vivo* settings (e.g., kidney imaging), while Baranger *et al.* provide an adaptive singular-vector-similarity rank rule for ultrafast Doppler [12]. Our benchmark set includes both literature families directly and a tile-local hybrid that combines Song-style overlap-add localization with a Baranger-style adaptive rank rule. Many open fUS releases do not preserve raw high-PRF complex IQ suitable for labeled Doppler-side detector evaluation, so we use labeled k-Wave scenarios and structurally labeled phantoms for quantitative ROC claims and use *in vivo* data for robustness and translation audits. Our contribution differs in where the adaptation enters and in the discipline used to test it: Yu-, Baranger-, and Song-style methods adapt the *clutter filter* itself and then typically read out power Doppler or a closely related conventional score, whereas we keep the residual cube fixed and change the *detector head*. Specifically, we evaluate a small family of downstream heads on identical residuals, including deliberately simple ablations such as whitened total power without Doppler band partition and a fixed band-limited ratio without whitening. This lets us test whether the gain comes from matched-subspace geometry, whitening

alone, or their combination. The result is a downstream detector-head framework for post-residual fUS, together with a strict-tail evaluation discipline, rather than another local SVD clutter filter.

Historical radar connection. Classical MVDR/LCMV/STAP and knowledge-aided variants motivate the broader idea of covariance-aware detection under structured interference [52–55, 60, 61, 64]. Here that influence is narrower and ultrasound-specific: local covariance whitening is an optional post-residual specialist mode, and the optional side-information layer is implemented only as a shrink-only, fail-safe score regularizer.

ULM pipelines and motion correction. Ultrafast ultrasound localization microscopy (ULM) achieves deep super-resolution vascular imaging by localizing sparse microbubble echoes over high-PRF ensembles [39]. Robust clutter suppression and motion correction are central to ULM pipelines, including subwavelength motion correction tailored to ultrafast acquisitions [40]. We do not claim improved microbubble localization performance here; rather, we use a public kHz IQ ULM benchmark dataset to report label-free stability and motion robustness of Doppler-side score maps under controlled motion corruption, illustrating how a tail-stable detector can complement upstream motion correction and downstream localization stages.

Twinkling artifact and Doppler phantoms. The Doppler "twinkling artifact" has been documented as a persistent color-flow phenomenon on calcifications and other strong reflectors, and is widely used as a diagnostic benchmark for Doppler processing chains. Leonov *et al.* provide an open-access database of raw beamformer signals with phantoms including a flow-through Gammex phantom and calculi/metal/plastic objects [42]. We use these data to (i) obtain a structurally labeled phantom ROC via B-mode-only tube segmentation and (ii) construct an artifact-heavy regularizer-hygiene regime where the optional penalty layer is expected to be conservative and auditable.

fUS detection and hemodynamic analysis. GLM-based detection and atlas-driven ROI analysis are common in hemodynamic fUS [47, 48]. In this paper we use those analyses as downstream readouts and translation audits; extending the detector to a fully dedicated hemodynamic scoring variant is future work rather than a claimed contribution here.

2.2 Baseline clutter suppression and score conventions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to the detector family.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define a conventional power Doppler (PD) brightness map used only as a display/proxy map for deterministic preprocessing (e.g. proxy masks) and visualization,

$$\mathrm{PD}^{\uparrow}(y, x) = \frac{1}{T} \sum_{t=0}^{T-1} |X_{\text{base}}(t, y, x)|^2 .$$

For quantitative evaluation and thresholding we use *flow-evidence* score maps where higher values indicate stronger evidence of flow, and we calibrate thresholds on the right tail of these score distributions. We denote the baseline ROC score map as $S_{\text{base}}(y, x)$ (power Doppler on X_{base}). For

Symbol / name	What it measures	Used for	Data requirement
S_{base}	baseline ROC score map on X_{base} (PD power; higher = more flow-like)	baseline ROC / low-FPR tables	IQ
$S_{\text{stap,pre}}$	pre-regularization matched-subspace detector score map	detector ROC / low-FPR tables	IQ + local covariance
$S_{\text{stap,post}}$	post-regularization shrink-only score map $wS_{\text{stap,pre}}$	regularizer behavior audits	IQ + feature map
$\text{PD}^{\uparrow}$ (display PD), masks	PD display/proxy map and derived masks	visualization and deterministic preprocessing (gating/protected sets)	IQ (or PD-only proxy mode)

Table 3: Detector-score and display-map taxonomy used throughout. ROC score maps are kept separate from display/proxy maps used only for visualization and deterministic preprocessing. Display/proxy maps are never used to compute ROC/TPR/FPR (only to define gating/protected-set masks, which are audited separately).

short-ensemble regimes we additionally report auxiliary, conventional baseline comparators (e.g. log-power Doppler and Kasai lag-1 magnitude [15]) to avoid baseline-score idiosyncrasies dominating conclusions (Notation). Note: $\text{PD}^{\uparrow}$ and S_{base} use the same underlying power statistic (up to fixed scaling); we keep separate names to make it explicit when PD is used only as a display/proxy map (masks/visuals) versus as a ROC score map.

2.3 Score definitions

Unless explicitly stated, the paper discusses three pre-regularization detector variants built from the same band-limited score family: a fixed matched-subspace head without whitening, a prespecified adaptive variant that promotes selected pixels onto a whitened branch when guard-band evidence indicates clutter, and a fully whitened covariance-adaptive variant. To keep the notation manageable, the remainder of the paper refers to these simply as the fixed, adaptive, and fully whitened detector modes. The shrink-only regularizer is analyzed separately as an optional post-processing layer.

3 Method

What changes relative to SVD + PD. Relative to a conventional ultrafast Doppler pipeline, the proposed detector family changes the final scoring stage rather than replacing upstream clutter suppression. After forming a residual slow-time signal, we score its energy inside a prespecified flow subspace rather than using only aggregate Doppler power. The simplest fixed head uses that score without local whitening, a prespecified adaptive variant promotes clutter-heavy pixels onto a whitened branch only when guard-band evidence warrants it, and a fully whitened variant is retained for regimes where stronger local adaptation is beneficial. The goal is not simply to increase vessel conspicuity, but to reduce nuisance-driven high scores in operating regions where false positives matter.

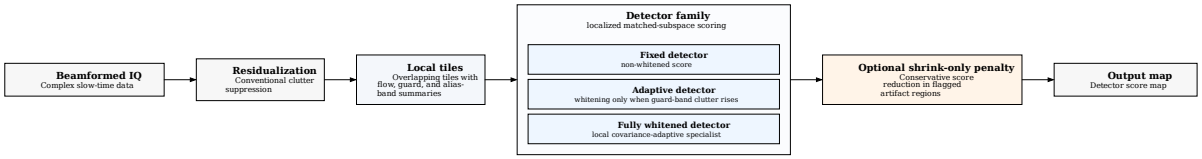

Figure 3: Method overview. Beamformed IQ first undergoes ordinary residualization, after which the detector operates tilewise in slow time. The fixed detector uses the band-limited matched-subspace score everywhere. The adaptive detector uses that same score by default and switches to a whitened branch only in tiles with strong guard-band evidence of localized clutter. The fully whitened detector applies whitening everywhere. An optional shrink-only penalty layer can then downweight high-risk regions while leaving protected vessel-proxy regions unchanged.

Plain-language interpretation. Readers do not need radar background to follow the method. The fixed head can be read simply as a local band-limited flow detector applied after ordinary residualization. The adaptive branch adds tile-local whitening only where guard-band energy indicates localized clutter, and the optional penalty layer acts as a conservative vessel-proxy masking rule that can only shrink scores. We retain a few historical abbreviations because they are compact, but in this paper they refer only to ultrasound-specific post-beamformed operations: "STAP" denotes optional tile-local slow-time whitening, "CFAR" denotes empirical negative-tail threshold calibration rather than an analytic scanner-time false-alarm law, and "KA" denotes a side-information regularizer that can suppress high-risk tiles but never amplify them. These are shorthand labels, not a literal import of classical radar processing blocks.

Pipeline overview.

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y, x)$.
2. For each tile Ω_i: form slow-time snapshots $\{x_{i,k}\}_{k=1}^{K}$ via spatial sampling and, optionally, a sliding-window temporal embedding (Hankel matrix construction), estimate conditioned covariance R_i, and either keep the raw snapshots or whiten them as $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute a self-normalized matched-subspace ratio score on the chosen tile representation, e.g. $S_{\text{stap,pre}}(i) = \frac{1}{K} \sum_k \frac{y_{i,k}^{\text{H}} P_f \, y_{i,k}}{y_{i,k}^{\text{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\text{H}} y_{i,k} + \epsilon}$, and aggregate over overlapping tiles to obtain $S_{\text{stap,pre}}(y, x)$. Setting $R_i = I$ recovers the simpler band-limited matched-subspace fixed head.
4. If the adaptive variant is used, retain the fixed head by default and promote pixels onto the whitened branch only where a prespecified guard-band clutter feature exceeds a fixed threshold.
5. If the optional penalty layer is enabled: compute a deterministic, shrink-only weight $w(i) \in (0, 1]$ from auxiliary band-energy statistics and vessel-proxy support features (details in the Supplementary Material), preserve a protected do-not-penalize set (scores unchanged), and output $S_{\text{stap,post}}(y, x) = w \, S_{\text{stap,pre}}(y, x)$. Else, $S_{\text{stap,post}}(y, x) = S_{\text{stap,pre}}(y, x)$.

In practical terms, the method asks a local question after ordinary clutter suppression: within a small spatial neighborhood, does the remaining slow-time activity look more like microvascular flow or more like locally structured nuisance? A detector-core ablation on phantom data is provided in Supplementary Material.

Three detector variants. Let $S_0(i)$ denote the non-whitened band-limited matched-subspace score on tile i and let $S_w(i)$ denote the fully whitened robust matched-subspace score computed from the same band geometry. The paper studies three concrete detector variants. The fixed detector uses $S_0(i)$ everywhere. The prespecified adaptive detector uses

$$S(i) = \begin{cases} S_w(i), & r_g^{\text{base}}(i) \geq \tau_g, \\ S_0(i), & r_g^{\text{base}}(i) < \tau_g, \end{cases}$$

where $r_g^{\text{base}}(i)$ is a baseline guard-band energy fraction and τ_g is fixed once and then held fixed across all reported experiments. The fully whitened detector uses $S_w(i)$ everywhere. In words, the adaptive detector keeps the simpler score in clean tiles and only turns on full whitening where the residual itself shows guard-band evidence of localized clutter.

3.1 Localized detector core and Doppler band geometry

Tile-based detector family. To mitigate the degraded tail sensitivity of global low-rank filters in high-clutter regimes, we study a localized matched-subspace detector family on overlapping spatial tiles. Its primary member keeps the same flow-band score without local whitening ($R_i = I$); its adaptive member uses that same score by default and promotes selected tiles onto a whitened branch; its fully whitened member adds a covariance-adaptive slow-time core in which spatial localization enters through tile-local covariance estimation. Let Ω_i denote tile i and L_t the slow-time aperture. From X_{base} we construct a set of slow-time training snapshots $\{x_{i,k} \in \mathbb{C}^{L_t}\}_{k=1}^{K}$ for each tile via spatial sampling and, when applicable, a sliding-window temporal embedding (Hankel construction) in slow time. Here K denotes the number of training snapshots used to estimate the tile covariance; in the fully Hankelized case, each pixel contributes $N = T - L_t + 1$ snapshots and pooling across Ω_i yields $K = N|\Omega_i|$ (or a subsampled/capped variant). We estimate a local covariance

$$\widehat{R}_i := \frac{1}{K} \sum_{k=1}^{K} x_{i,k} x_{i,k}^{\text{H}},$$

apply conditioning/diagonal loading to obtain R_i, and whiten snapshots as $y_{i,k} := R_i^{-1/2} x_{i,k}$.

Training support and Hankel dependence. In the fully Hankelized case, each pixel in Ω_i contributes $N = T - L_t + 1$ overlapping slow-time snapshots; pooling across the $|\Omega_i|$ pixels yields $K = N|\Omega_i|$ tile-local training vectors. These Hankelized snapshots are temporally correlated due to overlap, so K should not be interpreted as an independent-sample count. Our robust covariance estimators do not require independence, but the effective support can be significantly smaller than K; we therefore optionally subsample Hankel columns (stride) and cap the number of columns used for training to bound computation and avoid overweighting highly correlated samples (fixed per regime; Supplementary Material).

Robust covariance estimation and conditioning for the fully whitened mode. Whenever the fully whitened member of the detector family is used, we whiten with a regularized covariance R_i rather than the raw SCM $\widehat{R}_i$. The prespecified stress-test tables in this paper use a Tyler-type

M-estimator on the pooled tile snapshots, iterating

$$R_i^{(n+1)} \leftarrow \frac{L_t}{K} \sum_{k=1}^{K} \frac{x_{i,k} x_{i,k}^{\mathrm{H}}}{x_{i,k}^{\mathrm{H}} (R_i^{(n)})^{-1} x_{i,k}}, \qquad R_i^{(n+1)} \leftarrow \frac{L_t}{\mathrm{tr}(R_i^{(n+1)})} R_i^{(n+1)},$$

with at most 25 iterations and early stopping when $\|R^{(n+1)} - R^{(n)}\|_F / \|R^{(n)}\|_F < 10^{-4}$ (latency-only experiments may cap iterations; Supplementary Material). We then apply deterministic conditioning (shrinkage toward μI and diagonal loading $\lambda \mu I$, where $\mu = \mathrm{tr}(R)/L_t$ and λ is fixed per regime) to enforce target condition numbers. In very short-ensemble regimes we may fall back to a Huber IRLS covariance estimate for numerical stability (Supplementary Material). When the whitened member of the detector family is used, we use a single robust recipe based on Huber covariance estimation with trimmed high-energy training support and stronger loading. We refer to that prespecified recipe as the *Huber-whitened specialist*; exact numerical settings remain in the reproducibility appendix so the main text can focus on which member of the family is preferred in which regime.

Reader's guide. The core method in this paper is the fixed band-limited matched-subspace detector. The adaptive extension keeps that score by default and only turns on local whitening where guard-band clutter evidence is high. The fully whitened specialist applies local covariance whitening everywhere and is used only when that extra complexity is justified. The optional shrink-only regularizer is secondary to the main detector comparison and can only downweight scores. We mention STAP only as a historical influence: unlike classical radar STAP, this method operates after beamforming in slow time and does not use array-channel steering.

Figure 4 illustrates the local tile extraction and optional sliding-window temporal embedding (Hankel construction) used to form $\widehat{R}_i$ and contrasts it with the global Casorati stacking used by MC–SVD.

Band operators and matched-subspace detector score. We partition the Doppler spectrum into clutter/DC, flow, guard, and alias bands using projectors (P_o, P_f, P_g, P_a) (Notation). For short ensembles, these are exact, orthogonal DFT-bin projectors with $P_o + P_f + P_g + P_a = I$; for longer apertures they are leakage-controlled operators that are approximately orthogonal and are treated as band-energy statistics. We define a matched-subspace detector statistic by flow-subspace energy on whitened snapshots, e.g.,

$$S_{\mathrm{stap,pre}}(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{y_{i,k}^{\mathrm{H}} P_f \, y_{i,k}}{y_{i,k}^{\mathrm{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\mathrm{H}} y_{i,k} + \epsilon},$$

and form a pixel-level score map $S_{\mathrm{stap,pre}}(y, x)$ by overlap-add aggregation of the contributing tile statistics. This pre-regularization score is the primary detector output used for quantitative reporting unless explicitly stated otherwise.

Regularization and self-normalization avoidance. If we whitened using the exact sample covariance formed from the same snapshots being scored, the statistic collapses: the whitened sample covariance becomes (approximately) identity and averaged subspace energies concentrate near

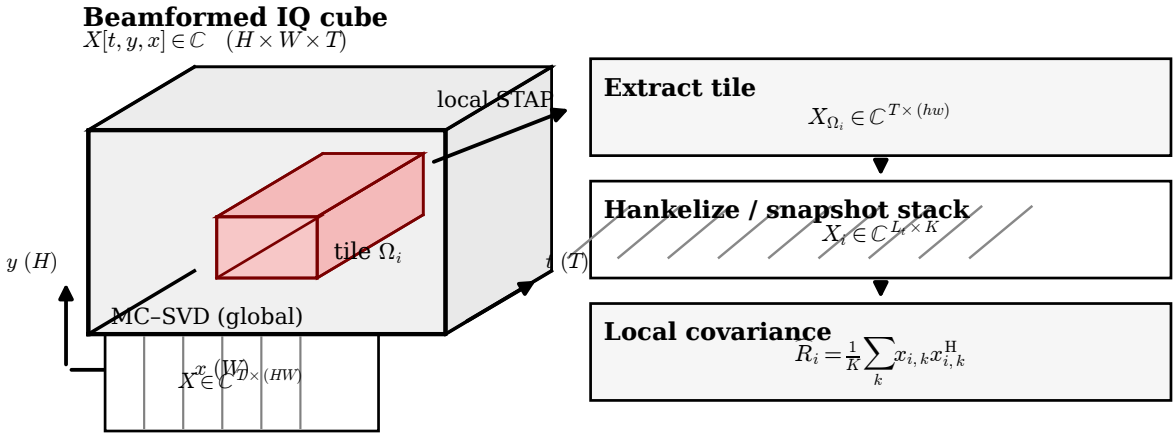

Figure 4: Schematic of sliding-window temporal embedding (Hankel matrix construction) and local covariance estimation. A localized spatial tile Ω_i is extracted from the beamformed slow-time data cube $X[t, y, x] \in \mathbb{C}$ $(H \times W \times T)$. Overlapping slow-time snapshots of length L_t are stacked (optionally Hankelized) to estimate the localized covariance matrix $\widehat{R}_i$ used only by the advanced whitened mode. Unlike global low-rank filters (e.g. MC–SVD) that operate on a global Casorati matrix, this localized formulation preserves spatial-spectral heterogeneity of microvascular flow and highly structured tissue clutter.

constants (e.g. $\mathrm{tr}(P_f)$), destroying discrimination. We avoid this by whitening with a *regularized* covariance R_i (robust estimation plus shrinkage/diagonal loading), so that typically $R_i \neq \widehat{R}_i$ and the matched-subspace energy remains informative. We do not rely on an analytic null distribution for threshold setting; operating points are empirically calibrated on labeled H_0 sets under the prespecified protocol of Section 4.1.

Matched-subspace statistic as a whitened subspace detector. Fix tile i and consider a single slow-time snapshot $x \in \mathbb{C}^{L_t}$. Under a Gaussian clutter-plus-noise model,

$$H_0 : \ x \sim \mathcal{CN}(0, R_i), \qquad H_1 : \ x = A_f s + n, \quad s \sim \mathcal{CN}(0, \sigma_s^2 I), \quad n \sim \mathcal{CN}(0, R_i),$$

where the columns of A_f span the flow subspace (equivalently the range of P_f). After whitening $y = R_i^{-1/2} x$, a matched-subspace detector based on whitened flow-subspace energy arises naturally under this colored-Gaussian model:

$$T(x) = x^{\mathrm{H}} R_i^{-1/2} P_f R_i^{-1/2} x = y^{\mathrm{H}} P_f \, y,$$

which is the standard whitened subspace-energy statistic for this setting. [62, 63] In implementation we report a self-normalized matched-subspace ratio that normalizes flow-subspace energy by the orthogonal complement (and a small ridge). It is motivated by the same whitened subspace geometry as $T(x)$, but it is not identical to the basic GLRT energy statistic; we use it because it behaves better numerically and yields more stable thresholding in practice. This self-normalized statistic is the pre-regularization detector score $S_{\mathrm{stap,pre}}(i)$.

3.2 Optional shrink-only regularization (brief main-text summary)

The main detector results in this paper are all carried by the pre-regularization detector heads above. We nevertheless study a bounded optional shrink-only regularizer because structured artifacts can still dominate extreme-tail operating points in some regimes. The mapping is deterministic, conservative, and uses auxiliary band-energy and proxy-support statistics only as *side information*; proxies are never used to compute ROC/TPR/FPR (only to define protected/candidate sets and compute masks, which are audited separately). Full construction details and safety contracts are left to the Supplementary Material.

Regularizer at a glance.

- **When it activates:** Only when deterministic sanity/support checks pass (valid band discretization; non-flat risk features; sufficient proxy support). Otherwise it is inert ($w \equiv 1$).
- **What it changes:** Applies a bounded, shrink-only weight $w(i) \in (0, 1]$ on an explicitly defined candidate set while preserving a protected do-not-penalize set (scores unchanged); the post-regularization score is $S_{\text{stap,post}} = w\, S_{\text{stap,pre}}$.
- **What it cannot do:** It cannot increase any pixel score (no amplification) and proxies are not used for ROC/TPR/FPR reporting; any apparent operating-point uplift reflects background-tail compression under shrink-only mapping.
- **How it fails safely:** If out of regime, it defaults to "off" ($w \equiv 1$) and logs an explicit reason label, reducing to the pre-regularization detector.

Core regularizer features. The optional shrink-only regularizer is driven by a small set of deterministic side-information features computed from the same tile-local slow-time vector z_i and Doppler-band projectors (P_o, P_f, P_g, P_a) used by the detector. We define

$$E_f(i) = z_i^{\mathrm{H}} P_f z_i, \qquad E_g(i) = z_i^{\mathrm{H}} P_g z_i, \qquad E_a(i) = z_i^{\mathrm{H}} P_a z_i,$$

with total in-band energy $E_{\text{tot}}(i) = z_i^{\mathrm{H}} z_i + \epsilon$. The main regularizer risk statistic is the alias-to-flow ratio

$$m_{\text{alias}}(i) = \log \frac{E_a(i) + \epsilon}{E_f(i) + \epsilon},$$

which is large when a tile is dominated by aliased or high-velocity nuisance rather than the prescribed flow band. We also monitor the guard-band fraction

$$r_g(i) = \frac{E_g(i)}{E_{\text{tot}}(i)},$$

and a deterministic flow-support proxy $c_f(i) \in [0, 1]$ to define conservative background-like and flow-like proxy sets

$$\mathcal{T}_{\text{bg}} := \{i : c_f(i) \le c_{\text{bg}}\}, \qquad \mathcal{T}_{\text{flow}} := \{i : c_f(i) \ge c_{\text{flow}}\}.$$

These quantities are used only to decide whether shrinkage is warranted and where it may be applied; they are not used as class labels for ROC reporting.

Shrink-only mapping (guarantee). We implement the optional penalty layer via a bounded weight $w(i) \in (0, 1]$ applied in score space:

$$S_{\text{stap,post}}(i) = w(i)\, S_{\text{stap,pre}}(i), \qquad w(i) \in (0, 1].$$

The Supplementary Material gives the full construction of $w(i)$, including protected sets, regime logic, deterministic sentinels, and the reciprocal scale-map convention used for some exported audit files.

Lemma 1 (Shrink-only monotonicity)**.** *Let $w(i) \in (0, 1]$ and define $S_{\text{stap,post}}(i) = w(i)\, S_{\text{stap,pre}}(i)$. For any threshold τ,*

$$\{S_{\text{stap,post}} \geq \tau\} \subseteq \{S_{\text{stap,pre}} \geq \tau\}.$$

Consequently, on any subset A (e.g. H_0), the tail probability cannot increase at fixed τ:

$$\mathbb{P}\{S_{\text{stap,post}} \geq \tau \mid A\} \leq \mathbb{P}\{S_{\text{stap,pre}} \geq \tau \mid A\}.$$

Proof. If $S_{\text{stap,post}}(i) \geq \tau$, then $S_{\text{stap,pre}}(i) = S_{\text{stap,post}}(i)/w(i) \geq \tau$ since $w(i) \leq 1$. □

When reporting strict low-FPR operating points where τ is reselected after regularization to satisfy $\mathbb{P}\{S_{\text{stap,post}} \geq \tau \mid H_0\} = \alpha$, changes in TPR reflect background-tail compression despite scores never increasing.

3.3 Algorithmic integrity and deterministic safeguards

To make the architecture auditable and to avoid leakage or discretization artifacts masquerading as performance gains, we enforce deterministic, label-free safeguards:

- **Conditional execution as a decision-rule variant.** To reduce latency, local whitening can be skipped outside a flow-proxy region and replaced by the baseline score there. Because this changes the overall decision rule, any conditional-execution reporting is treated as a distinct algorithmic variant and is accompanied by leakage controls (same-window vs. disjoint-window vs. randomized proxy masks). Unless otherwise stated, conditional execution uses *disjoint-window* proxy gating for detection reporting.
- **Short-ensemble band discretization sanity.** For extremely short ensembles (e.g. $N \in \{5, 9, 17\}$), the coarse frequency resolution $\Delta f = \text{PRF}/N$ can cause requested bands to collapse or overlap after discretization. If the discretized bands are empty, DC-only, or overlapping, the optional shrink-only regularizer is disabled ($w \equiv 1$) with an explicit reason label.
- **Proxy-mask circularity control.** Proxy maps are never used to compute ROC/TPR/FPR; they are used only to define compute masks (conditional execution) or protected/candidate sets for the optional regularizer, which are audited separately. Quantitative ROC claims are restricted to regimes with independent H_1/H_0 labels (simulator truth or B-mode structural geometry).

For auditability, we log explicit reason labels so that "regularizer off" ($w \equiv 1$) is an expected success mode under flat or non-actionable evidence. Supplementary Material provides modality-agnostic sufficient conditions and additional sentinels used to interpret when the optional penalty layer should be active or inert.

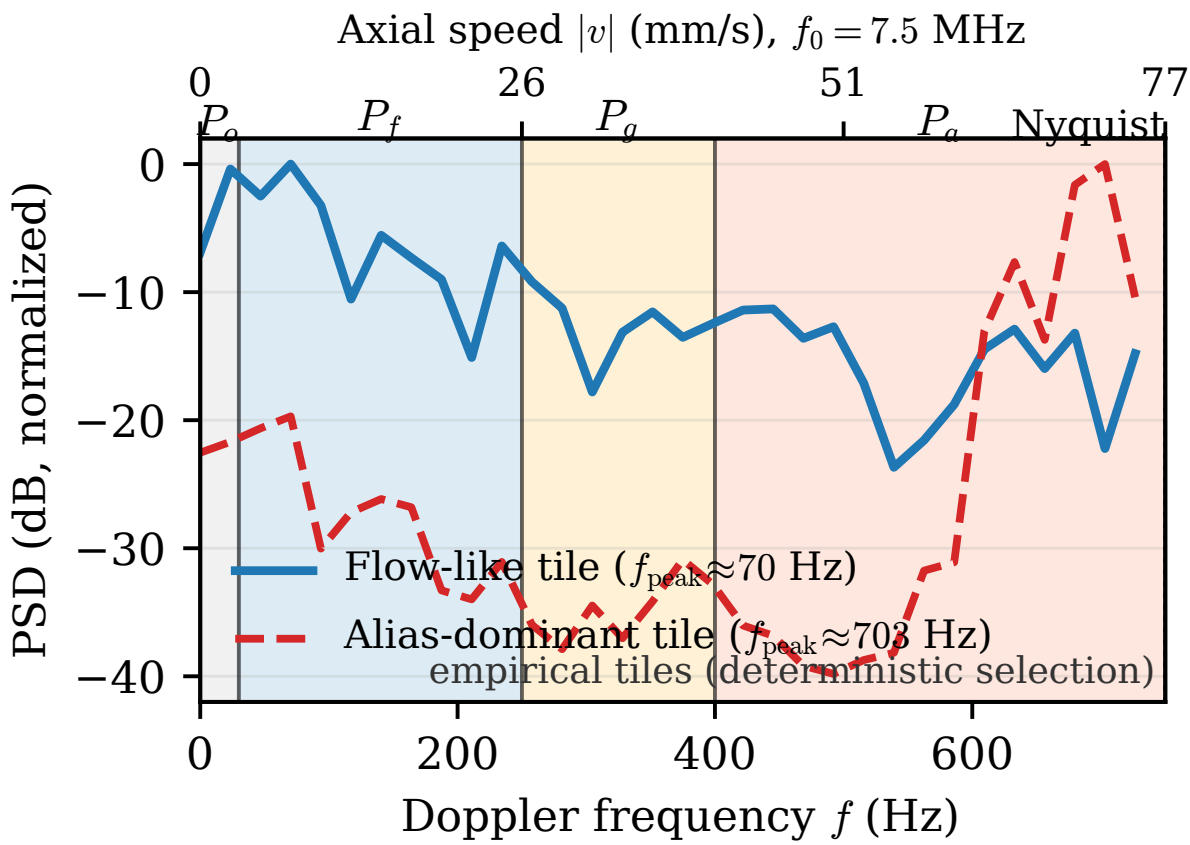

Figure 5: Doppler power spectral density (PSD) partitioning and kinematic band geometry (empirical example). Curves show one-sided PSDs of the tile-mean slow-time signal on the alias-dominant labeled brain simulation (PRF = 1500 Hz): a flow-like tile (peak in P_f) and an alias-dominant tile (peak in P_a). The Nyquist interval is deterministically partitioned into DC/high-amplitude clutter (P_o), microvascular flow (P_f), a transition guard band (P_g), and aliased high-velocity artifacts or vibration (P_a). Auxiliary band-energy descriptors (e.g. alias-to-flow ratios) are used for interpretation and, optionally, for shrink-only tail regularization (Supplementary Material).

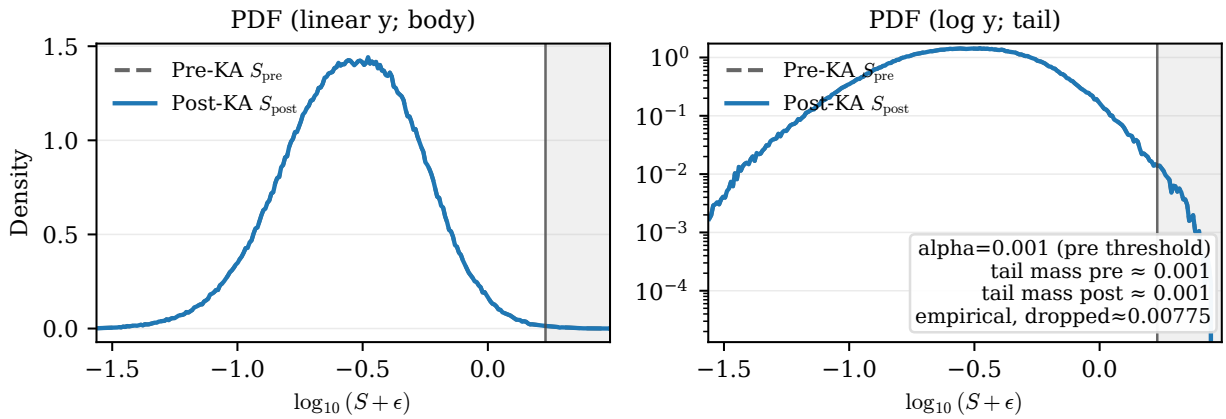

Figure 6: Empirical statistical effect of the strictly bounded, shrink-only penalty on a heavy-tailed background-proxy score distribution from an artificial-calculi acquisition in the open Doppler Twinkling Artifact database. The matched-subspace score before the optional penalty exhibits a heavy right tail characteristic of structured artifacts. Applying the non-increasing weight $w(i) \in (0, 1]$ yields the post-penalty score $S_{\text{stap,post}} = w(i)S_{\text{stap,pre}}$, compressing the background tail while preserving an explicit do-not-penalize protected set (scores unchanged on high-confidence flow locations). The vertical line indicates the pre-penalty threshold τ_α chosen so that $\mathbb{P}\{S_{\text{stap,pre}} \geq \tau_\alpha \mid \text{bg proxy}\} = \alpha$ (here $\alpha = 10^{-3}$); the shaded region highlights the corresponding right tail, and the post-penalty tail mass is evaluated at the same τ_α.

4 Experimental Design

4.1 Prespecified operating protocol and operating-point reporting

All processing settings (tile geometry, aperture, band definitions, conditioning targets, and prior thresholds) are fixed for each regime before held-out results are reported. Detection is evaluated by thresholding the high-score tail (right tail) at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\mathrm{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on regime). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points. We emphasize $\alpha = 10^{-3}$–10^{-4} because those operating points expose nuisance-tail failure modes cleanly, not because routine fUS use must literally operate at those exact values. The same map-trustworthiness problem is already visible at a conventional matched background FPR of 10^{-2} in the worked Gammex example of Figure 2, where the baseline thresholded map is dominated by off-lumen detections while the matched-subspace score recovers lumen-consistent structure.

Threshold calibration and deployment. Unless otherwise stated, τ_α is calibrated per window (or per cine sequence) on that unit's negative set to report $\mathrm{TPR}(\alpha)$ at the intended operating point; in labeled regimes, the negative set is available for *exact* calibration, while in deployment it would be estimated via a negative-only training procedure or from a fixed calibration set. This per-window protocol should therefore not be read as requiring oracle per-window tuning in deployment. To keep that distinction explicit, the Results section now includes two non-retrospective transfer checks in the main reader path: a fixed-calibration audit on the held-out SIMUS/PyMUST benchmark and a complementary block-split fixed-calibration audit on the ULM structural audit.

How the prespecified defaults were chosen. The numerical settings in this pipeline do not all play the same role, and they were not selected in the same way. Band edges are tied to acquisition physics: PRF fixes the discrete Doppler grid and Nyquist limit, while the flow/guard/alias partitions are chosen to separate slow parenchymal microvascular content from transition-band and aliased high-velocity nuisance on that grid. Tile geometry and L_t are locality/support choices: they are chosen once per regime to preserve spatial adaptation while retaining enough slow-time support for stable covariance estimation, then held fixed. Conditioning parameters (for the fully whitened mode) are chosen to keep the robust covariance well behaved on development audits rather than to maximize held-out task scores. The adaptive threshold τ_g is a single frozen guard-energy cutoff chosen to preserve the fixed head in cleaner regimes while turning on whitening only when guard-band clutter evidence is clearly non-negligible. The regularizer proxy thresholds are deliberately coarse rather than finely optimized: $c_{\mathrm{bg}} = 0.05$ means "essentially background-like tile coverage," $c_{\mathrm{flow}} = 0.20$ means "substantial flow-support coverage," $q_{\mathrm{risk}} = 0.90$ defines a conservative upper-tail risk cutoff, and the p_{shrink} bounds are safety sentinels on how much of the candidate set may be penalized. A supplementary stability audit perturbs the finalized τ_g and c_{flow} thresholds by $\pm 10\%$ and shows that the detector family does not qualitatively collapse under those changes (Supplementary Material).

4.2 Integrity checks and deterministic sentinels

This work is a reproducible processing pipeline (decode → baseline → detector score → optional shrink-only penalty layer). Several practical pathologies can invalidate tail-rate reporting or spuriously activate the penalty layer (e.g. missing acquisition metadata, coarse short-ensemble band discretization, proxy-mask circularity, or compute-gating leakage). We therefore enforce deterministic, label-free integrity checks that either (i) record provenance and proceed conservatively or (ii) disable the penalty layer ($w \equiv 1$) with an explicit reason label. A concise checklist (and where each safeguard is logged) is provided in Supplementary Material.

4.3 Regimes and claim scope

Claims are stratified by regime so that quantitative detection results remain clearly separated from label-free robustness or translation audits. Table 4 also makes the role of open datasets explicit, because the real-data story in this paper is intentionally bounded but not singular.

- **Labeled brain simulation scenarios (k-Wave).** These are controlled stress tests for strict-tail behavior. Across the open-skull, alias-dominant, and structured-clutter scenarios, the matched-subspace detector family retains measurable sensitivity at ultra-low FPR, with the strongest gains in the alias-dominant and structured-clutter settings.
- **Phantom acquisitions (structurally labeled Gammex flow phantom).** These provide independent structure-based labels on real scanner data. We derive B-mode-only tube masks, compare conventional short-ensemble Doppler scores with matched-subspace detector-family scores, and report structural low-FPR operating points and within-ensemble motion ladders.
- **Prespecified held-out SIMUS/PyMUST benchmark.** This is the main held-out use-case-motivated synthetic benchmark in the paper. It uses the SIMUS/PyMUST forward model, but the evaluated settings are narrowed using real-data fUS use cases and observed operating characteristics rather than generic simulator coverage alone [8, 9, 34, 45–47]. We report held-out structural and functional performance for the prespecified mobile setting and intra-operative parenchymal setting, together with a held-out stress-frontier confirmation on the mobile setting under cardiac-like pulsation and short ensembles.
- **ULM Zenodo 7883227 (real kHz IQ).** This dataset is used for motion-robustness analyses on real IQ, with controlled rigid and nonrigid motion injection and explicit fairness checks. We also report a bounded same-acquisition structural audit using the published PALA super-resolution vascular rendering registered back onto the IQ grid, with a vessel-core versus perivascular-shell endpoint rather than a functional-task label.
- **Shin rat-brain real-IQ sequence.** This open real-IQ dataset provides a second label-free motion-stress evaluation under a prespecified low-frequency operating setting. It is used as corroborating real-IQ evidence rather than as a task ROC benchmark.
- **Supplementary analyses.** Additional label-free audits (e.g. PD-only whole-brain fUS) and diagnostic panels are provided in Supplementary Material.

Table 4 also makes the current validation ladder explicit. This paper covers simulation, phantom, and one bounded same-acquisition structural audit on open real IQ, but the next rung is only now becoming public and remains fragmented rather than standardized. Recent resources include simultaneous fUS–Neuropixels recordings in awake mice with open superior-colliculus fUS and electrophysiology data, raw fUS plus two-photon multimodal platforms in awake mice, and bedside or human task-fUS releases through craniectomy or sonolucent cranial windows [8, 22–25].

Regime	Open-data status / data type	Independent labels?	What it supports
Labeled brain simulation scenarios (k-Wave)	project-generated simulated IQ	yes (simulator truth)	controlled strict-tail mechanism and latency stress tests
Twinkling database: Gammex + calculi	open real scanner phantom Doppler IQ	Gammex: yes (B-mode structure); calculi: no	structural ROC on real data, phantom motion ladders, and regularizer hygiene
Prespecified held-out SIMUS/PyMUST benchmark	open SIMUS/PyMUST forward model with real-data anchoring	yes (simulated label masks)	main held-out structural and functional benchmark, held-out stress-frontier confirmation under cardiac-like pulsation and short ensembles, and separate-bank fixed-calibration transfer on held-out seeds
ULM Zenodo 7883227	open real kHz IQ	bounded structural labels from a published localization-derived vascular reference	real-IQ motion robustness, same-residual score audit, a bounded structural-label vascular-fidelity audit on real in-vivo brain IQ, and complementary block-split fixed-calibration transfer
Shin rat-brain IQ sequence	open real kHz IQ	no independent task labels	second real-IQ corroboration under controlled residual-motion stress

Table 4: Claim-support map, canonical benchmark names, and validation ladder used throughout the paper. Quantitative detection performance is reported across three labeled tiers: brain simulation stress tests, the structurally labeled Gammex phantom, and one bounded same-acquisition ULM structural audit using a published localization-derived vascular reference. The held-out SIMUS/PyMUST benchmark provides the main held-out stress-test tier. The other real-IQ sections remain explicitly bounded as robustness or corroborative audits rather than as broad task ROC benchmarks.

Those advances are highly encouraging, but they still do not yet constitute a single public benchmark that combines open slow-time data, exact task timing, an independent biological anchor, and repeated sessions suitable for threshold-transfer evaluation.

5 Results

Results roadmap. We begin with the held-out SIMUS/PyMUST benchmark, but we lead that section with the harder held-out stress frontier rather than with the cleaner reference checkpoint alone. This makes the main structural story less vulnerable to the objection that the task is too easy: the detector family is first shown under clinically grounded pulsation and short-ensemble pressure, and the cleaner held-out checkpoint is then retained to show which fixed head should be treated as the default. We then report the supporting brain simulation stress tests at ultra-low FPR, the Gammex phantom structural results, and finally the open real-IQ evaluations: the ULM

Regime (sampling)	Baseline	Detector head (prespecified settings)	Doppler bands (Hz)
Labeled brain simulation scenarios (k-Wave; PRF 1500; 64-frame windows)	MC–SVD (energy fraction $e = 0.90$); adaptive SVD (SV similarity cutoff); block-wise local SVD (overlap-add; $e = 0.90$); RPCA; HOSVD (baselines scored by PD on residual)	tiles 8×8, stride 3; $L_t = 8$; robust covariance (Tyler-type) with diagonal loading $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [30, 250]$, guard $[250, 400]$, $P_a = [400, 750]$
Gammex flow phantom (linear17; PRF 2500; $N = 17$ shots; pooled over cine frames)	baseline SVD band-pass keep $[2, 17]$; conventional short-ensemble comparators (PD/log-PD/Kasai)	tiles 8×8; $L_t = 16$; Tyler-type covariance with $\lambda = 0.07$; stride 6 (along) / 4 (across); matched-subspace flow-band score	$P_f = [150, 450]$, guard $[450, 700]$, $P_a = [700, 1200]$
ULM Zenodo 7883227 motion and same-residual audits (PRF 1000; 128-frame blocks)	MC–SVD baseline with energy fraction $e = 0.975$ chosen by a one-time label-free sweep	tiles 8×8, stride 3; $L_t = 64$; SCM covariance with $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [10, 150]$, guard $[150, 200]$, $P_a = [200, 500]$

Table 5: Prespecified operating settings for the main regimes. All settings are fixed per regime prior to reporting results; exact commands and exported metadata are provided in Supplementary Material. The bounded ULM structural audit uses a separate prespecified 64-frame no-registration specialist configuration described explicitly in Section 5.5.

motion audit, the bounded ULM structural audit, and the Shin real-IQ audit. Additional dataset evaluations and diagnostic panels are provided in Supplementary Material.

5.1 Held-out SIMUS/PyMUST benchmark

This moving-scatterer benchmark is intentionally narrower than a generic SIMUS parameter sweep. It retains the SIMUS/PyMUST moving-scatterer forward model [45, 46], but the evaluated settings are chosen to reflect whole-brain, cranial-window, mobile, and intra-operative fUS use cases rather than generic simulator realism alone [8, 9, 34, 47]. We evaluate two prespecified settings: a mobile setting and an intra-operative parenchymal setting. Both were calibrated against real-data operating characteristics and then fixed before held-out testing. A third, more surface-dominated intra-operative setting remains exploratory and is not used for quantitative claims. In reviewer-safe terms, this is a use-case-motivated synthetic benchmark with prespecified nuisance settings informed by reported fUS operating conditions; it does not substitute for prospective clinical validation.

One version-control point is worth stating explicitly. During an earlier SIMUS motion-ladder audit we found a motion-scaling bug in the moving-scatterer perturbation path: some simulated motion amplitudes were mis-scaled, which affected the absolute difficulty of auxiliary motion-stress runs. After fixing that bug, we refroze the reader-facing benchmark reported here. The SIMUS results in this paper therefore always refer to the corrected held-out SIMUS/PyMUST benchmark, its held-out stress frontier, and its fixed-calibration transfer audit. Earlier development variants are retained only in the Supplementary Material as fairness checkpoints and should not be used when interpreting the main claims.

For transparency, this benchmark is defined by explicit held-out splits and explicit label masks rather than by post hoc example selection. The structural checkpoint uses development seeds 125–126 to choose the final method families and settings, and held-out evaluation seeds 127–128 for

the results in Table 7. The functional checkpoint uses a disjoint development split (seeds 221–222) and held-out evaluation split (seeds 223–224) under a common downstream readout. Structurally, each evaluated case exports the masks `mask_h1_pf_main`, `mask_h0_bg`, `mask_h0_nuisance_pa`, and `mask_h1_alias_qc`; the evaluation script computes the summary metrics directly from those masks under a fixed right-tail score convention.

For the structural task, the positive class is unaliased parenchymal Pf-band flow and the negative classes are (i) ordinary background and (ii) explicit nuisance structures such as superficial high-velocity or specular components. The three summary metrics used throughout this section are therefore plain-language clinical surrogates: $AUC_{\text{main/bg}}$ measures separation of target flow from ordinary background, $AUC_{\text{main/nuis}}$ measures separation of target flow from clinically relevant nuisance, and FPR_{nuis} @ $TPR_{\text{main}} = 0.5$ asks how many nuisance detections survive when the detector is tuned to retain half of the target positives. This last quantity is the most direct nuisance-tail metric in the table.

For the functional task, we use the same prespecified settings and a common background-controlled GLM-style readout on simulated task/null runs. Two comparisons are reported because they answer different questions: end-to-end pipeline ranking asks which full processing chain performs best after all residualizer-setting interactions, whereas the same-residual audit asks whether the matched-subspace detector improves the *same* residual cube under the same downstream readout.

Because the cleaner held-out checkpoint can otherwise look deceptively easy, we lead the structural SIMUS/PyMUST results with the harder held-out stress frontier on the same mobile scene family. These rows use the same benchmark infrastructure but push it toward two clinically grounded nuisance regimes: cardiac-like low-frequency tissue pulsation and progressively shorter slow-time support. The pulsation rows use a fixed 1.30 Hz cardiac-like nuisance term whose amplitude is increased over reference, moderate, and hard settings; this is intended to probe the same low-frequency tissue-motion regime emphasized by Demené *et al.* as critical for SVD clutter filtering and by Baranger *et al.* in neonatal/open-skull Doppler settings [11, 12]. The short-ensemble rows reduce the support from 64 frames to 48 and 32 frames to mimic clinically constrained acquisitions in which shorter windows are used to limit motion corruption or workflow burden [1, 22, 34]. For tractability, this confirmation fixes attention to the residualizer families that dominated the preliminary reduced-grid search (RPCA and adaptive-global SVD) and then reruns the held-out evaluation at the full reporting tier.

Table 6 is the harder headline structural result. On every held-out stressed row, the strongest detector-family stack outperforms the strongest public comparator evaluated on that same stressed row. The public reference on these rows is not a cleaner operating point but the best conventional stack under the same perturbation, which keeps the comparison honest. The row-wise winner is not uniform, and that is scientifically useful: the simpler matched-subspace default already wins the cardiac reference row and the shortest-ensemble row, whereas the fully whitened specialist takes the more aggressive pulsation rows and the intermediate short-ensemble row. This is the clearest evidence in the paper for why a detector family is needed at all rather than a single universal score. Table 7 should therefore be read as the cleaner default-selection checkpoint rather than as the only headline structural result. The first two metric columns still measure separability, while the last metric column directly measures nuisance leakage at a matched main-signal sensitivity operating point. The most important pattern in this checkpoint is that both "Adaptive Global SVD → matched-subspace default" and "RPCA → matched-subspace default" drive the nuisance-leakage

Cardiac-like pulsation

Level / perturbation	Best detector-family stack	Metrics (Public → Detector)
Ref.	RPCA	$AUC_{main/nuis}$: 0.381 → 0.879
1.0x cardiac amplitude at 1.30 Hz	→ Fixed matched-subspace	FPR_{nuis}@0.5: 0.743 → 0.034
Mod.	Adaptive-global SVD	$AUC_{main/nuis}$: 0.388 → 0.883
2.0x cardiac amplitude at 1.30 Hz	→ Whitened specialist	FPR_{nuis}@0.5: 0.668 → 0.037
Hard	Adaptive-global SVD	$AUC_{main/nuis}$: 0.399 → 0.882
3.0x cardiac amplitude at 1.30 Hz	→ Whitened specialist	FPR_{nuis}@0.5: 0.609 → 0.025

Short ensemble

Level / perturbation	Best detector-family stack	Metrics (Public → Detector)
Ref.	RPCA	$AUC_{main/nuis}$: 0.381 → 0.879
64 frames (42.7 ms)	→ Fixed matched-subspace	FPR_{nuis}@0.5: 0.743 → 0.034
Mod.	Adaptive-global SVD	$AUC_{main/nuis}$: 0.390 → 0.841
48 frames (32.0 ms)	→ Whitened specialist	FPR_{nuis}@0.5: 0.754 → 0.056
Hard	RPCA	$AUC_{main/nuis}$: 0.426 → 0.772
32 frames (21.3 ms)	→ Fixed matched-subspace	FPR_{nuis}@0.5: 0.617 → 0.166

Table 6: Held-out SIMUS mobile stress frontier. Each row compares the best detector-family stack with the strongest public comparator on the same held-out evaluation seed. On every row in this reduced table, the strongest public comparator was RPCA → PD. The cleaner structural checkpoint is shown separately to identify the fixed default detector head.

column to 0.000, even though the upstream residualizers are different. That common result is why we interpret the matched-subspace head itself, rather than only the choice of residualizer, as the main driver of the nuisance-tail gain on this benchmark. The checkpoint is scientifically useful because it identifies the simpler band-limited matched-subspace head as the strongest fixed default on held-out structural data, while the harder rows in Table 6 show where the adaptive and fully whitened variants add value in clutter-heavier regimes.

For translational readers, the key question is not whether the method makes vessels look better on average, but whether fewer nuisance detections survive when the detector is tuned to preserve clinically relevant parenchymal flow. That is the practical meaning of FPR_{nuis} @ $TPR_{\text{main}} = 0.5$: it asks how much structured false signal remains at a matched target-flow sensitivity operating point. Because fUS is already being used in intra-operative, bedside, cranial-window, and emerging mobile settings [1, 8, 20–22], that nuisance-leakage tradeoff is tied directly to usability and reader trust rather than to image aesthetics alone.

That per-window operating-point protocol is still an evaluation protocol, not a scanner-time thresholding rule. To address the more practical question directly, we also ran a fixed-calibration transfer check in which the threshold is learned once from separate calibration seeds 125–126 and then applied unchanged to the held-out evaluation seeds 127–128. Table 8 reports that result at nominal $\alpha = 10^{-3}$. On the mobile setting, the fixed matched-subspace head preserves nonzero held-out main recall (0.004 [0.002, 0.006]) while lowering realized nuisance FPR from 0.016 [0.015, 0.018] for PD to 0.003 [0.000, 0.006]. On the intra-operative parenchymal setting, both methods remain near the nominal background target, but only the fixed head suppresses nuisance leakage to 0.000 [0.000, 0.000], whereas PD remains at 0.489 [0.464, 0.514]. This is not the full prospective scanner workflow yet, but it does show that the fixed default selected by the cleaner checkpoint survives a separate-bank threshold transfer on the benchmark that anchors the paper.

Processing chain	Total runtime (s)	$\mathrm{AUC}_{\mathrm{main/bg}}$	$\mathrm{AUC}_{\mathrm{main/nuis}}$	$\mathrm{FPR}_{\mathrm{nuis}}$ at $\mathrm{TPR}_{\mathrm{main}} = 0.5$
HOSVD → Kasai	0.167	0.753	0.561	0.453
Adaptive Global SVD → matched-subspace default	0.638	0.851	0.953	0.000
RPCA → Kasai	1.783	0.837	0.442	0.593
RPCA → matched-subspace default	2.407	0.994	1.000	0.000

Table 7: Cleaner prespecified held-out SIMUS/PyMUST structural checkpoint on held-out evaluation seeds under the prespecified evaluation setting. We highlight two conventional references and the two strongest fixed-head detector-family chains. $AUC_{\mathrm{main/bg}}$ is the area under the ROC curve for separating main signal from background. $AUC_{\mathrm{main/nuis}}$ is the area under the ROC curve for separating main signal from nuisance-labeled pixels. FPR_{nuis} @ $TPR_{\mathrm{main}} = 0.5$ is the nuisance false-positive rate at the threshold where main-signal true-positive rate equals 0.5. The key reader-facing pattern is that the two rows sharing the same downstream matched-subspace head both drive FPR_{nuis} @ $TPR_{\mathrm{main}} = 0.5$ to 0.000 despite using different upstream residualizers. This checkpoint is retained because it identifies the simpler band-limited matched-subspace detector as the fixed structural default; the harder stress-frontier rows in Table 6 show where the adaptive and fully whitened extensions earn their extra complexity.

Method	Mobile			Intra-operative parenchymal		
	$\mathrm{TPR}_{\mathrm{main}}$	realized $\mathrm{FPR}_{\mathrm{bg}}$	realized $\mathrm{FPR}_{\mathrm{nuis}}$	$\mathrm{TPR}_{\mathrm{main}}$	realized $\mathrm{FPR}_{\mathrm{bg}}$	realized $\mathrm{FPR}_{\mathrm{nuis}}$
Power Doppler on MC–SVD residual	0.000 [0.000,0.000]	0.002 [0.000,0.003]	0.016 [0.015,0.018]	0.012 [0.006,0.017]	0.001 [0.001,0.001]	0.489 [0.464,0.514]
Fixed matched-subspace head	0.004 [0.002,0.006]	0.002 [0.001,0.003]	0.003 [0.000,0.006]	0.006 [0.004,0.008]	0.001 [0.001,0.001]	0.000 [0.000,0.000]

Table 8: Fixed-calibration transfer on the prespecified held-out SIMUS/PyMUST structural benchmark at nominal $\alpha = 10^{-3}$. For each setting, a single threshold is learned once from pooled background negatives on separate calibration seeds 125–126 and then applied unchanged to the held-out evaluation seeds 127–128. Entries report mean [min,max] across the two held-out seeds, so this table is a direct non-retrospective threshold-transfer check rather than the per-window oracle calibration used for ROC reporting elsewhere.

The stress-frontier table should also prevent the wrong reading of the cleaner checkpoint. The near-perfect nuisance numbers in Table 7 are not the paper's only structural evidence, and they should not be interpreted as meaning that the benchmark is universally trivial. Once the same scene family is pushed toward pulsatile tissue leakage and shorter support, the nuisance-control task becomes visibly harder and the detector-family hierarchy becomes easier to interpret. That is why the main paper now keeps both views of the benchmark together: the harder stress frontier to show nontrivial held-out difficulty, and the cleaner checkpoint to show which fixed head should actually be deployed by default.

Functional same-residual detector-head comparison. To test whether the fully whitened variant adds value beyond upstream residualization in function-priority settings, we report a compact same-residual comparison on the prespecified functional benchmark settings. For each fixed residual stream, we compare the fully whitened detector to conventional Doppler-style score heads using the same held-out data and the same downstream GLM-style readout.

Table 9 makes the functional story more explicit than a simple win-count. In the strict held-out same-residual audit, the fully whitened detector is positive in 5 of 6 residual families in both settings, with median held-out selection-score gains of +0.065 [0.032, 0.091] for the intra-operative

Benchmark setting	Best end-to-end chain	Matched-subspace-positive families	Median Δ selection score [IQR]	Median $\Delta\mathrm{AUC}_{\mathrm{act/nuis}}$ (IQR)
Intra-operative parenchymal	RPCA $\rightarrow$ PD after matched-subspace	5/6	+0.065 [+0.032,+0.091]	+0.026 [+0.014,+0.052]
Mobile	HOSVD $\rightarrow$ PD	5/6	+0.111 [+0.032,+0.142]	+0.059 [+0.011,+0.066]

Table 9: Held-out functional same-residual detector-head summary on the prespecified SIMUS/PyMUST settings using the common readout `bgcdf_outside_glm`. "Best end-to-end chain" is the strongest full held-out processing chain on that setting. "Matched-subspace-positive families" counts how many of the six prespecified residual families favored the matched-subspace detector stage over the native conventional score in the stricter held-out same-residual comparison. "PD after matched-subspace" denotes the paper's prespecified downstream PD-style readout after the matched-subspace stage; see Methods for the exact implementation. Δ selection score and $\Delta\mathrm{AUC}_{\mathrm{act/nuis}}$ are reported as matched-subspace minus native, summarized by the median and interquartile range across those six families.

parenchymal setting and +0.111 [0.032, 0.142] for the mobile setting. The corresponding median gains in activation-versus-nuisance AUC are +0.026 [0.014, 0.052] and +0.059 [0.011, 0.066]. Those are not large enough to carry the paper by themselves, but they are directionally consistent and quantitatively nontrivial under a common readout. A broader bounded targeted-search follow-up remains positive in 11 of 12 family/setting comparisons and is kept in the Supplementary Material rather than in the main reader path.

Within the detector family itself, the robustly whitened Huber variant therefore remains worth keeping even though it is not the structural default. This is one prespecified whitening recipe carried across the functional, phantom, and labeled-stress-test regimes rather than a different score retuned for each dataset. That cross-regime consistency is the main reason we retain it as a distinct fully whitened variant inside the matched-subspace detector framework. The adaptive variant described later sits between this fully whitened head and the simple fixed default: it does not replace the functional fully whitened result, but it provides a cleaner translational compromise than forcing full whitening everywhere.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on labeled brain simulations, structurally labeled phantom regimes (Gammex flow phantom sequences; labels derived from B-mode geometry), and the prespecified held-out SIMUS/PyMUST benchmark using the strict low-FPR operating protocol of Section 4.1. Open real-IQ datasets are used primarily for bounded score-audit and motion-robustness evidence rather than for task ROC; the ULM section additionally includes a bounded localization-derived structural audit on real brain IQ, which we treat as an in-vivo vascular-fidelity check rather than as a decisive task-efficacy benchmark. Conditional execution is a decision-rule variant that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Supplementary Material.

5.2 Brain simulation stress tests

This subsection is intentionally supportive rather than primary. The SIMUS/PyMUST benchmark above is the main translational result in the paper. Here we turn to simpler labeled brain simulations only to isolate detector behavior under controlled nuisance stress tests. We report labeled low-FPR operating points on these scenarios using flow-evidence scores under a right-tail convention (higher = more flow-like; thresholds calibrated on the high-score tail): a baseline power Doppler score and the proposed matched-subspace score (Notation). Baseline methods compare clutter suppression strategies (MC–SVD; adaptive SVD with a spatial singular-vector similarity cutoff; block-wise local SVD with overlap-add; RPCA; HOSVD) and are scored using the same power Doppler statistic; the matched-subspace detector family is evaluated on top of the MC–SVD residual. Across these scenarios, the PD-style baselines degrade sharply in the ultra-low-FPR tail, while the localized band-limited matched-subspace score retains measurable sensitivity at ultra-low FPR (i.e., $\mathrm{TPR}(\alpha) > 0$ at $\alpha = 10^{-4}$–10^{-3}), including in the structured-clutter setting where nuisance deliberately leaks into the positive class. Because these tables are meant to explain mechanism rather than headline performance, the main text keeps only the two more clinically interpretable simulation settings: an open-skull surface-artifact stress test and a structured-clutter leakage stress test. A third alias-dominant positive control showed the same qualitative trends but is omitted here for readability. Tables 10–12 should therefore be read as component audits on labeled stress tests, not as the primary translational result. Importantly, the MC–SVD+PD baseline row and the matched-subspace row share the identical MC–SVD residual cube and differ only in the downstream score (PD versus matched-subspace), isolating the effect of scoring rather than the clutter-suppression stage. This is why the detector-component and detector-swap tables belong here: they explain *why* the main SIMUS/PyMUST benchmark later favors certain processing chains, but they do not replace that benchmark. A supplementary threshold-transfer audit on the labeled brain simulation stress tests still shows that the robustly whitened variant remains close to target in both the open-skull and structured-clutter settings. Those simulation-only checks are now secondary to the stronger deployment-transfer results reported in the main text on the held-out SIMUS/PyMUST benchmark and on the ULM structural audit.

Explicit limitations from the labeled-brain audits. Two points matter here. First, full whitening is not the structural default: on the prespecified held-out structural benchmark, the simpler band-limited matched-subspace detector remains stronger on nuisance-control metrics. Second, the robustly whitened variant is stable rather than brittle on these labeled-brain stress tests: open-skull realized FPR stays near target under both pairwise cross-window transfer (median 1.10×10^{-4}, $q_{75} = 1.64 \times 10^{-4}$ at $\alpha = 10^{-4}$) and a leave-one-window-out fixed calibration bank (median 1.20×10^{-4}), and the operating-point results are numerically unchanged across covariance-training trim $q \in \{0.05, 0.08, 0.10\}$. The remaining limitation is therefore about where whitening earns its extra complexity, not about broad instability of the fully whitened recipe itself.

Baseline fairness and prespecified settings. All baseline hyperparameters are fixed prior to reporting results. For the labeled brain simulation scenarios, the MC–SVD energy-fraction cutoff is selected once and then held at $e = 0.90$ for all reported windows. The comparator families span global low-rank filtering (MC–SVD and adaptive-global SVD), block-wise local filtering (fixed-energy and adaptive-local SVD), RPCA, and HOSVD. Baseline rows are then evaluated with the same right-tail Doppler score so that differences reflect the residualizer rather than a change in

Method	Open-skull surface-artifact stress test			Structured-clutter leakage stress test		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
PD on MC–SVD residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Whitened power (no band)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Unwhitened ratio (no whitening)	0.8689 (0.8674,0.8689)	0.8947 (0.8831,0.8985)	0.9389 (0.9166,0.9400)	0.9104 (0.8900,0.9189)	0.9373 (0.9058,0.9389)	0.9619 (0.9512,0.9689)
Fully whitened matched-subspace detector	0.0730 (0.0008,0.0865)	0.1288 (0.0012,0.2222)	0.2218 (0.0015,0.4368)	0.5871 (0.5006,0.6682)	0.7205 (0.6048,0.7785)	0.8731 (0.8189,0.8758)

Table 10: Detector ablations on labeled brain simulation stress tests (MC–SVD residual, prespecified settings). All rows share the identical MC–SVD residual cube and differ only in the detector statistic: baseline PD, total power after tile-local whitening (no Doppler band partition), matched-subspace ratio without whitening (R=I), and the fully whitened matched-subspace detector. The two column groups correspond to an open-skull surface-artifact stress test and a structured-clutter leakage stress test. Numbers are medians (IQR) over five disjoint 64-frame windows (offsets 0/64/128/192/256).

the downstream scoring rule. We do *not* retune these baseline families on the held-out windows to maximize their own strict-tail TPR after seeing the test cases; the main SIMUS/PyMUST benchmark instead gives each family a development-split configuration-selection step before held-out reporting, while the labeled brain stress tests use fixed representative settings chosen once and then frozen. The compact family summary is provided in Supplementary Material; in the main text we keep the emphasis on what the comparisons mean scientifically rather than on implementation taxonomy.

Detector-component ablation (whitening versus band selectivity). To address the concern that the proposed score is merely "local whitening plus a ratio" under a new name, we report two deliberately simple detector ablations on the identical MC–SVD residual cube: (i) *whitened total power* (apply $R_i^{-1/2}$ but do not partition the Doppler spectrum) and (ii) an *unwhitened* matched-subspace ratio (set $R_i = I$ but retain the fixed flow-band subspace). Table 10 is the key mechanism check. Whitening alone without band partition stays at zero across the reported strict-tail operating points, while the unwhitened band-limited ratio remains strong in both labeled stress tests. In these labeled brain simulations, most of the gain therefore comes from the detector geometry itself rather than from whitening alone. We interpret the contribution accordingly: the novelty here is downstream detector-head design and evaluation discipline, with local covariance adaptation as one useful extension rather than the whole story.

Detector-swap composability audit (matched-subspace scoring on competing residuals). To isolate detector effects from the clutter-suppression stage, we also compute the matched-subspace detector score on residuals produced by competing clutter filters (RPCA and HOSVD) and compare it to power Doppler scoring on the same residual under the identical window/mask protocol. Table 11 reports this paired-residual audit. Table 11 should not be read as contradicting Table 7. The labeled simulation detector-swap audit isolates detector-head behavior on the controlled k-Wave stress tests under fixed residual streams, whereas Table 7 reports end-to-end performance on the prespecified held-out structural benchmark. These answer different questions: detector-head fairness versus end-to-end deployment tradeoff. Under the prespecified settings used throughout, strict-tail improvements do not automatically persist when composing the matched-subspace detector with more aggressive low-rank baselines. We therefore treat it as a second-stage detector defined for a coarse baseline residual (MC–SVD in the main brain-simulation pipeline),

Method	Open-skull surface-artifact stress test			Structured-clutter leakage stress test		
	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$	$\alpha = 10^{-4}$	$\alpha = 3\times10^{-4}$	$\alpha = 10^{-3}$
PD on RPCA residual	0.0000 (0.0000,0.0046)	0.0015 (0.0000,0.0088)	0.0042 (0.0042,0.0173)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Matched-subspace detector on RPCA residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
PD on HOSVD residual	0.0000 (0.0000,0.0015)	0.0015 (0.0015,0.0015)	0.0046 (0.0046,0.0077)	0.0000 (0.0000,0.0015)	0.0000 (0.0000,0.0015)	0.0015 (0.0000,0.0031)
Matched-subspace detector on HOSVD residual	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0004)	0.0019 (0.0004,0.0019)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)

Table 11: Detector-swap fairness check on labeled brain simulation stress tests: PD vs matched-subspace detector scores computed on the *same* residual produced by a competing clutter filter (paired bundles). The two column groups correspond to an open-skull surface-artifact stress test and a structured-clutter leakage stress test. Numbers are medians (IQR) over five disjoint 64-frame windows (offsets 0/64/128/192/256).

Trim setting	Open-skull stress test			Structured-clutter stress test		
	Within TPR@10^{-4}	Cross FPR med. (q75)	Fixed-cal FPR med. (q75)	Within TPR@10^{-4}	Cross FPR med. (q75)	Fixed-cal FPR med. (q75)
$q = 0.05$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)
$q = 0.08$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)
$q = 0.10$	0.852	1.10e-04 (1.64e-04)	1.20e-04 (1.39e-04)	0.917	9.96e-05 (2.44e-04)	7.97e-05 (1.99e-04)

Table 12: Robust-whitening stability audit for the prespecified fully whitened detector on the labeled brain stress tests. The only changing parameter is the high-energy covariance-training trim q. Entries report within-window TPR at the strictest operating point ($\alpha = 10^{-4}$) together with realized FPR under pairwise cross-window transfer and a leave-one-window-out fixed calibration bank. The reported values are numerically unchanged across $q \in \{0.05, 0.08, 0.10\}$, indicating that the finalized advanced recipe is stable to modest covariance-training trim variation in these stress tests.

and use the controlled detector swap on the same MC–SVD residual as the primary isolation of detector effects.

Robust-whitening stability audit (labeled brain simulations). Because the whitened detector relies on tile-local covariance estimation, a natural concern is sensitivity to contaminated training support and threshold calibration. Table 12 examines those concerns directly for the prespecified Huber-whitened recipe by varying only the high-energy covariance-training trim over $q \in \{0.05, 0.08, 0.10\}$ and reporting both within-window performance and transfer behavior at the strictest operating point. The result is intentionally simple: the labeled-brain operating-point results are numerically unchanged across those trim settings, and open-skull realized FPR remains close to target under both pairwise cross-window transfer and a fixed calibration bank. We therefore treat contamination-robust covariance training as part of the finalized whitened recipe rather than as an unresolved instability of the method family. We further report a 1D sensitivity sweep over covariance diagonal loading λ, tile geometry, and slow-time aperture L_t on the canonical open-skull pilot (Supplementary Material).

Relaxed operating-point sanity anchor (labeled brain simulations). Several baseline rows in the supplementary labeled-brain strict-tail table take the value $\mathrm{TPR}(\alpha) \approx 0$ at $\alpha \leq 10^{-3}$. To keep the main Results focused on the main SIMUS/PyMUST benchmark and the strongest labeled stress tests, we place the relaxed-operating-point sanity table ($\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$) in the Supplementary Material and retain here only the representative tail-collapse visualization that explains why those strict-tail zeros occur.

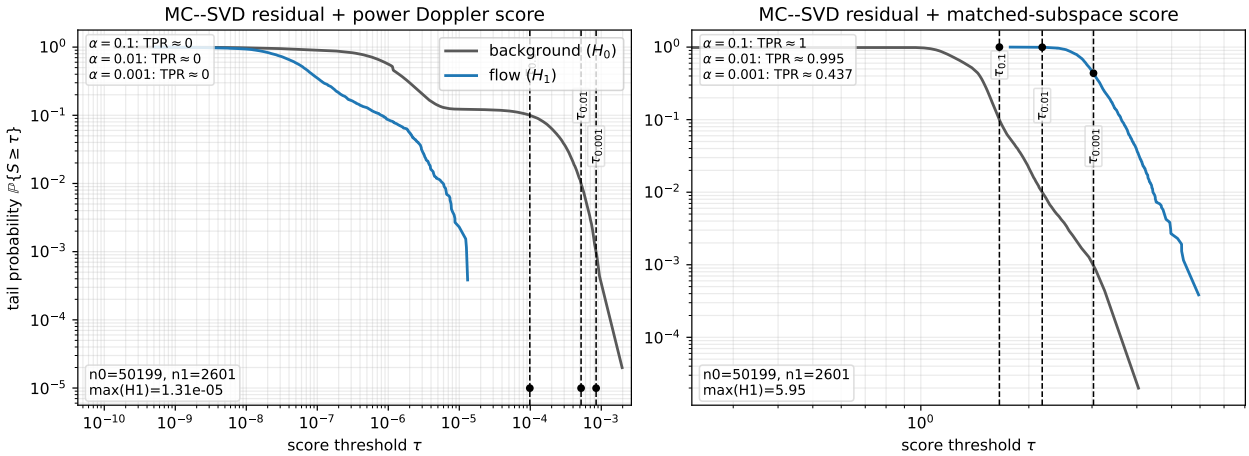

Figure 7: Representative open-skull labeled brain window illustrating strict-tail collapse of PD-after-filter baselines. Curves show empirical tail probabilities $\mathbb{P}\{S \geq \tau\}$ for background (H_0) and flow (H_1) score distributions versus score threshold τ, for (left) the baseline power Doppler score on the MC–SVD residual and (right) the matched-subspace score computed on the same residual before the optional shrink-only penalty. Vertical lines mark background-calibrated thresholds τ_α at $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and markers show the corresponding $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$. For the baseline PD score, $\tau_{10^{-3}}$ lies above $\max(H_1)$, yielding TPR ≈ 0 by construction; the matched-subspace score retains nonzero sensitivity at the same strict-tail operating point.

Why PD-after-filter baselines can collapse in the strict tail. A baseline clutter filter can yield nontrivial separation at moderate operating points yet still fail at ultra-low FPR if the background score distribution has an extreme right tail: the background-calibrated threshold τ_α (e.g. the 99.9th percentile of H_0 at $\alpha = 10^{-3}$) can lie above essentially all H_1 pixels, yielding $\mathrm{TPR}(\alpha) \approx 0$ by construction. For example, in a representative open-skull labeled brain window the MC–SVD+PD baseline background quantiles are $q_{0.99} \approx 5.9$ and $q_{0.999} \approx 9.3$ while the flow $q_{0.999} \approx 0.094$, so no flow pixels exceed the $\alpha = 10^{-3}$ threshold; Figure 7 visualizes the corresponding tail geometry ($\tau_{10^{-3}} > \max(H_1)$) for the baseline PD score and contrasts it with the matched-subspace score computed on the same residual. This tail-dominance failure mode motivates both the strict low-FPR operating discipline and detector heads that explicitly reshape the tail, beginning with band-limited matched-subspace scoring and, when needed, tile-local covariance whitening.

Latency and conditional execution. We also report representative runtimes for the final detector family evaluated in this manuscript rather than for an older whitening-heavy research configuration. In this evaluation, the labeled-brain stress tests use the simple unwhitened matched-subspace default with conditional execution, the Shin real-IQ evaluation uses the adaptive variant, and Gammex uses the short-ensemble raw-baseline plus unwhitened matched-subspace head. Conditional execution still changes the decision rule by deciding which tiles receive the adaptive detector; we therefore treat it as part of the reported detector configuration rather than as a cost-free acceleration of the same method. Unless otherwise stated, any conditional-execution reporting uses the same leakage-controlled, disjoint-window proxy discipline as the detection results.
Latency is measured by replaying multiple windows or frames in a single GPU process and reporting

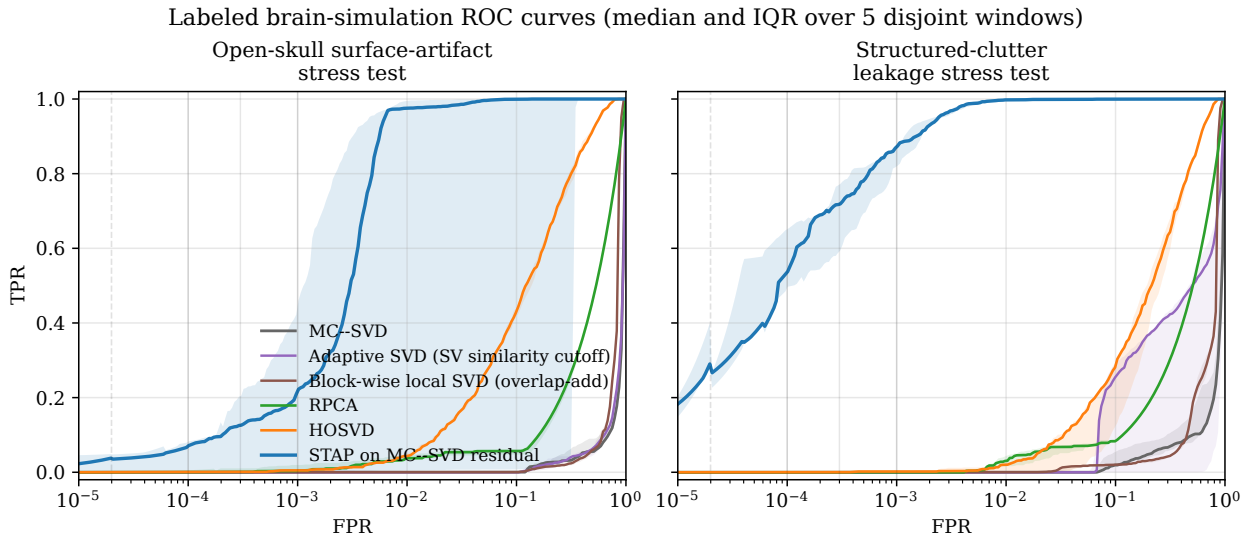

Figure 8: Controlled labeled-brain stress tests: ROC-style curves (TPR vs FPR on a log-x axis) under the same 64-frame window protocol used for the low-FPR operating-point reporting in the main text. Curves include MC–SVD, adaptive and local SVD baselines, RPCA, HOSVD, and the matched-subspace detector; each curve shows the median and interquartile range over five disjoint windows. For readability the main figure keeps the two more clinically interpretable simulation settings (open-skull surface artifact and structured-clutter leakage); the alias-dominant positive control is reported in the Supplementary Material. The moderate-FPR region illustrates that several PD-after-filter baselines can show separation at relaxed operating points while saturating at strict-tail operating points. The vertical dashed line indicates the supported floor $fpr_{\min} = 1/n_{\text{neg}}$.

steady-state means over windows/frames 2..N. The first replay window/frame is excluded because it includes one-time startup overheads. We time the baseline and detector stages separately (excluding disk I/O), and "Total" reports baseline+detector wall time. "Budget" denotes the acquired slow-time duration of the corresponding ensemble: $64/1500 \approx 42.7$ ms for the brain windows, $128/1000 = 128$ ms for Shin, and $17/2500 = 6.8$ ms for the Gammex cines. The reported real-time factor (RTF) is therefore Total/Budget.

Portable-workstation feasibility (tempered). A common critique of covariance-adaptive fUS scoring is that local covariance conditioning and the associated matrix factorizations are prohibitively expensive when applied densely across an image grid. Table 13 shows that, once the detector family is timed in the final reported configuration rather than in a whitening-heavy research setup, the picture is substantially better on an RTX 4080 SUPER. All three labeled-brain stress tests run below the full acquisition-PRF budget (RTF $\approx$ 0.86–0.88), the Shin rat-brain real-IQ evaluation also remains below budget (RTF $\approx$ 0.84), and both Gammex views now run below the 17-shot cine-frame budget (RTF $\approx$ 0.88 along-view and $\approx$ 0.64 across-view).[1] These results suggest that nuisance-aware matched-subspace scoring need not require server-class hardware, while still

[1]Latency is reported from steady-state replay (windows/frames 2..N) and excludes startup overhead and disk I/O. Table 13 reports the final detector family used in the manuscript rather than a single worst-case fully whitened configuration applied everywhere.

Scenario	Baseline (s)	Detector (s)	Total (s)	Budget (s)	RTF
Open-skull surface-artifact stress test (64-frame window)	0.024	0.012	0.037	0.043	0.86
Alias-dominant positive control (64-frame window)	0.025	0.012	0.038	0.043	0.88
Structured-clutter leakage stress test (64-frame window)	0.026	0.012	0.037	0.043	0.87
Shin rat-brain real-IQ evaluation (128-frame window)	0.006	0.102	0.108	0.128	0.84
Gammex flow phantom, along view (17-shot cine frame)	0.000	0.006	0.006	0.007	0.88
Gammex flow phantom, across view (17-shot cine frame)	0.000	0.004	0.004	0.007	0.64

Table 13: Steady-state latency summary on an NVIDIA GeForce RTX 4080 SUPER 16GB for the final detector family evaluated in this manuscript. Times are means over windows/frames 2..N, exclude disk I/O, and compare baseline and detector stages under the reported detector configuration rather than under an earlier whitening-heavy timing setup.

making clear that throughput depends on the chosen detector variant and should be interpreted with the explicit decision-rule caveats of conditional execution and adaptive whitening.
A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order 10^{-3}) at the target FPRs. We therefore treat conditional execution as a distinct decision rule rather than a cost-free speedup and place the full open-skull leakage table with bootstrap intervals in the Supplementary Material.

5.3 Gammex phantom: structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [42], which provides complex beamformer output from phantom acquisitions on a clinical scanner. Two parts of that dataset matter here for different reasons. The *Gammex flow phantom* provides a simple vessel-like geometry that lets us derive independent structure-based flow/background labels from B-mode anatomy alone. The *artificial calculi* sequences provide an artifact-heavy setting in which twinkling is expected and are therefore useful for auditing whether the shrink-only penalty layer behaves conservatively rather than spuriously inflating scores. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Leonov *et al.* explicitly report PRF for some acquisition families, including the transverse linear-probe Gammex acquisition, but also note that PRF information is not consistently preserved across the database. We therefore fix PRF once per sequence family, using the source-paper value where it is explicitly documented and otherwise recording a sequence-level override in bundle metadata. Band-discretization sanity checks disable the optional penalty ($w \equiv 1$) when discretized Pf/Pa bands are empty or overlapping (Section 3). Decoding/QC details are provided in Supplementary Material (PRF provenance and decoder sanity); reproduction commands and manifests are provided in Supplementary Material.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler or matched-subspace outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment; Figure 9 shows representative overlays. These structural labels support a defensible phantom ROC.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences used here ($N = 17$ shots; PRF values listed in Supplementary Material), we use a prespecified short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, detector aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are set to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Supplementary Material.

Structural-label low-FPR ROC (Gammex). At the strict FPR levels reported here, several conventional baselines collapse not because they are universally nonfunctional, but because residual nuisance structure dominates their upper score tail and exhausts the allowed false-positive budget before true flow targets are recovered. We therefore pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong twinkling/artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. Figure 10 visualizes this effect for a representative baseline comparator. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 14 summarizes a broader same-residual score comparison for the two Gammex views under those same settings: power Doppler and Kasai on the baseline residual, plus three detector-family score variants evaluated on that identical residual cube (whitened power, the fixed matched-subspace detector, and the fully whitened matched-subspace detector). For compactness we omit log-PD from the main table because it is monotone-equivalent to PD here and numerically identical at the reported operating points; the full pooled summaries remain available in Supplementary Material. The practical relevance is already visible at a conventional operating point. On the representative Figure 2 frame at matched background FPR 10^{-2}, the baseline thresholded map contains 884 suprathreshold pixels and none lie inside the structural lumen mask, whereas the matched-subspace score recovers 586 lumen pixels while holding the same background allowance. The stricter operating points reported below should

Method / score	Along (linear17 @ PRF 2500)			Across (linear17 @ PRF 2500)		
	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$	TPR@$\alpha = 10^{-4}$	TPR@$\alpha = 3\times10^{-4}$	TPR@$\alpha = 10^{-3}$
Baseline (power Doppler)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Baseline (Kasai lag-1 power)	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]	0.00000 [0.00000,0.00000]
Detector ablation (whitened power)	0.0012 [0.0011,0.0014]	0.0119 [0.0114,0.0124]	0.0532 [0.0521,0.0543]	0.4820 [0.4776,0.4863]	0.5155 [0.5120,0.5191]	0.5813 [0.5784,0.5842]
Detector ablation (unwhitened ratio)	0.00000 [0.00000,0.00001]	0.00000 [0.00000,0.00001]	0.00001 [0.00000,0.00002]	0.0574 [0.0534,0.0616]	0.0602 [0.0564,0.0645]	0.0653 [0.0612,0.0693]
Fully whitened matched-subspace detector	0.00000 [0.00000,0.00000]	0.00003 [0.00000,0.00007]	0.00031 [0.00020,0.00042]	0.0491 [0.0470,0.0514]	0.0998 [0.0970,0.1026]	0.2237 [0.2197,0.2280]

Table 14: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). This is a same-residual detector-head comparison: every row is evaluated on the identical baseline SVD band-pass residual cube, changing only the detector head. The baseline rows are power Doppler and Kasai lag-1 power on that residual; the detector ablations replace the fully whitened matched-subspace detector with either total whitened slow-time power (no Doppler band partition) or the same flow-band matched-subspace ratio without covariance whitening ($R = I$). Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\mathrm{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6 \times 10^{-7}$) and $n_{\mathrm{bg}} = 1.97 \times 10^6$ for the across view ($fpr_{\min} \approx 5.1 \times 10^{-7}$). Reproducibility details are provided in Supplementary Material.

therefore be read as a cleaner stress test of the same map-trustworthiness problem rather than as a claim that only 10^{-4} matters in practice. This decomposition clarifies the phantom fairness story. First, the baseline filter is not removing lumen flow: once the downstream score is changed on the same residual cube, both whitened-power and the fully whitened matched-subspace detector retain nonzero sensitivity. Second, local whitening is the decisive ingredient in this artifact-heavy phantom: removing whitening collapses the detector strongly on both views, especially the along view. Third, the full flow-band matched-subspace ratio is *not* the strongest score on Gammex: total whitened slow-time power is stronger on both views under this short-ensemble setting, while the matched-subspace ratio is more conservative. We therefore treat Gammex as a structurally labeled artifact-tail stress test that supports the value of local adaptive whitening and same-residual score analysis, not as proof that the matched-subspace ratio is universally the best choice in every phantom regime. The view asymmetry still matters: the across view provides the clearest structural-label phantom evidence, whereas the along view remains a harsher artifact-tail stress test under this short-ensemble profile. To bound the remaining fairness objections, we also keep a small untuned sanity audit in Supplementary Material: on the first 20 cine frames, applying the matched-subspace detector directly to raw registered IQ changes TPR@10^{-3} only from 3.48×10^{-4} to 2.61×10^{-4} on the along view and from 0.202 to 0.205 on the across view; perturbing the frozen along-view PRF from 2000 to 3000 Hz leaves TPR@10^{-3} unchanged at 3.48×10^{-4} in that limited audit. This phantom section should therefore still be read as an artifact-tail stress test under a prespecified short-ensemble profile, not as proof that every conventional head is universally zero under any plausible PRF or residualizer.

A compact matched-subspace-only ablation and PRF-override sensitivity check for Gammex are provided in the Supplementary Material. We keep them supplementary because they are profile-sanity diagnostics rather than the primary structural-label result.

Within-ensemble motion ladders (labeled, real acquisition). To obtain a labeled motion-robustness curve on a real acquisition without external motion ground truth, we inject synthetic *within-ensemble* motion directly into the CFM cube: for each cine frame we apply a per-shot rigid

Regime	Penalty active / inactive bundles	median ρ_{shrink}	background-tail rate at pre-penalty $\tau_{10^{-3}}$ (pre→post)	background-tail clusters at $\tau_{10^{-3}}$ (pre→post)
Twinkling calculi (PRF 500; $L_t = 8$)	50/0	0.102	$1.00 \times 10^{-3} \to 9.38 \times 10^{-4}$	$888 \to 832$

Table 15: Compact shrink-only penalty audit on the artifact-heavy Twinkling calculi regime. Under the prespecified short-ensemble profile, the shrink-only penalty is active on all 50 bundles (49 safety-only, 1 uplift-eligible), and the protected set is preserved exactly (maximum absolute score change = 0 on protected pixels). The fixed-threshold audit uses the pre-penalty threshold $\tau_{10^{-3}}$ chosen so that the background proxy satisfies $\mathbb{P}\{S_{\text{stap,pre}} \geq \tau_{10^{-3}}\} = 10^{-3}$; at that same threshold, the shrink-only penalty modestly compresses the background tail without score inflation. This is a quantitative hygiene audit, not a primary ROC claim. Extended region/reason panels and representative pre/post tail plots remain in Supplementary Material.

translation $(\Delta y_t, \Delta x_t)$ to the N CFM shots prior to baseline and matched-subspace scoring, with motion kinds `rw` (random walk) and `step`. Because the structural tube masks are derived from B-mode and fixed per frame, this produces a labeled "motion ladder" where the only change is within-ensemble corruption. We generate motion-corrupted ensembles (85 cine frames; amplitudes 0–2 px) and aggregate using frame-level bootstraps for CIs together with an explicit amp=0 equivalence check against the non-motion reference. Figure 12 shows the resulting curves; reproducibility details are provided in Supplementary Material.

5.4 Compact shrink-only penalty audit on artifact-heavy calculi

Because the main labeled detection tables report the matched-subspace score before the optional shrink-only penalty, we include one compact quantitative penalty audit in a regime where the regularizer is expected to act conservatively rather than to introduce a new headline efficacy claim. We use the artifact-heavy Twinkling calculi sequences under the prespecified short-ensemble setting and summarize activation, protected-set invariance, and fixed-threshold tail hygiene.

5.5 ULM motion audit (real kHz IQ)

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [37, 38] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Supplementary Material).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Supplementary Material).

Nonrigid motion injection and fairness alignment ablation. We evaluate motion robustness by injecting controlled motion directly into the beamformed IQ cube and re-running the baseline plus matched-subspace detector under a frozen ULM profile (PRF = 1000 Hz; $L_t = 64$; tiles

8×8; stride 3; SCM covariance; diagonal loading $\lambda = 0.07$; baseline MC–SVD with frozen $e = 0.975$). Because motion is injected at the beamformed IQ stage, this test stresses post-beamforming robustness to residual motion mismatch; it does not model channel-domain decorrelation or skull-induced phase-aberration physics. Motion ladders span two nonrigid families: `brainlike` (rigid random-walk translation plus a smooth, depth-decaying elastic warp with temporally correlated coefficients and micro-jitter) and `elastic` (nonrigid-only; rigid and micro-jitter disabled). For each amplitude we compute label-free robustness endpoints on the baseline score S_{base} and the matched-subspace score $S_{\text{stap,pre}}$ relative to the no-motion reference under three *common* alignment modes (none / common B-mode / injected GT proxy) plus a diagnostic per-method "self" alignment. We additionally report degeneracy checks (std ratios, top-k overlap, and tail quantile ratios) to rule out "stability by collapse." Figure 13 shows the main `brainlike` sweep; additional motion variants are provided in Supplementary Material.

Endpoint definition. For a score map $S_{\text{meth}}(a)$ at motion amplitude a and the corresponding no-motion reference $S_{\text{meth}}(0)$ (same block), we compute $\rho_{\text{meth}}(a)$ as the Pearson correlation between the aligned maps on a central crop Ω_a that excludes a margin of $\lceil a \rceil + 2$ pixels to avoid boundary artifacts from warping. Correlations are computed directly on the exported right-tail scores (no log transform) and use the same crop/alignment for baseline and matched-subspace maps in the "common" alignment modes.

Representative quantitative endpoints (ULM). At a 3 px motion amplitude under the injected GT alignment proxy, the median correlation on the baseline score S_{base} drops to ≈ 0.20 (`brainlike`), while the matched-subspace score $S_{\text{stap,pre}}$ remains ≈ 0.75 (medians over blocks 1–3; full tabular summaries and additional motion variants are provided in Supplementary Material). Degeneracy checks confirm that this stability is not explained by map collapse: the median std ratio and 0.999-quantile ratio of $S_{\text{stap,pre}}$ remain nontrivial across amplitudes, and top-k overlap decays gradually rather than saturating to trivial values.

ULM structural audit on the same dataset. To obtain a more anatomical structural readout on the same open real-IQ data, we use the published PALA rat-brain ULM example as an external vascular reference derived from microbubble localization microscopy rather than from the Doppler scores being evaluated [37, 38]. We first reproduce the corresponding short-horizon PALA PowDop rendering from the Zenodo IQ blocks using the published odd-block selection and SVD cutoff, register that published PowDop crop back to the Zenodo IQ field of view, and then downsample the published super-resolution MatOut vascular rendering onto the short-window evaluation lattice. From that registered localization map we derive a conservative vessel-core versus perivascular-shell label pair: the vessel mask is a peak-based core mask, and the background mask is the surrounding shell, so the endpoint asks whether a short-window detector concentrates score on the vascular core rather than the immediately adjacent parenchyma. That one-time PALA-to-IQ registration is part of the label construction only. For the short-window detector itself we keep frame-to-frame registration disabled in the frozen audit, because re-enabling it changes the structural ROC only below the fourth decimal place while adding substantial latency.

On this endpoint, the strongest specialist score is no longer PD or Kasai. In the completed 10-block / 20-window audit with 64-frame windows, the fully whitened matched-subspace specialist reaches AUC 0.526 [0.514, 0.536], compared with 0.503 [0.493, 0.513] for PD and 0.502 [0.492, 0.512] for Kasai. At matched 70% vessel-core recall, it also slightly lowers perivascular-shell false positives,

Score	core TPR	realized shell FPR
Baseline (power Doppler)	0.000 [0.000,0.000]	0.001 [0.000,0.002]
Baseline (Kasai lag-1 magnitude)	0.000 [0.000,0.000]	0.001 [0.000,0.002]
Whitened matched-subspace specialist	0.002 [0.000,0.004]	0.001 [0.000,0.002]

Table 16: Fixed-calibration transfer on the PALA-backed ULM structural audit at nominal $\alpha = 10^{-3}$. A threshold is learned once from pooled perivascular-shell negatives on one 5-block calibration bank and then applied unchanged to the complementary 5 held-out blocks; the reported values pool the two complementary block splits and show window-level means with 95% bootstrap CIs over held-out windows. This is therefore a genuine non-retrospective threshold-transfer check on real in-vivo IQ rather than per-block retuning.

from 0.701 [0.687, 0.715] for PD and 0.704 [0.689, 0.717] for Kasai to 0.690 [0.670, 0.710]. The effect is modest, but it is consistently positive across the audited windows. That result also fits the endpoint: the ULM structural audit rewards concentration on the localization-derived vascular core relative to the adjacent shell, whereas the held-out SIMUS/PyMUST benchmark rewards strong fixed downstream robustness against broader structured nuisance. On an RTX 4080 SUPER, the same 64-frame no-registration specialist configuration runs at 41.3 ms steady-state total for a 64 ms acquisition budget, so this audit is not tied to an offline-only implementation. The claim here is structural, not functional: the labels come from a published localization-derived vascular rendering on the same acquisition, not from an independent modality or a task-based ground truth.

Figure 1 gives the visual counterpart to this result by showing the localization-derived structural masks together with a representative same-window comparison between baseline power Doppler and the whitened matched-subspace specialist, while Table 2 gives the compact held-out quantitative summary already introduced near the front of the paper. To keep that table readable, it reports only the conventional PD-family and Kasai baselines together with the strongest specialist detector head, and omits secondary derived diagnostics such as detector-filtered PD. The figure is intended only to show how the two maps differ anatomically on a representative window.

A second deployment-oriented check uses the same structural endpoint but removes per-window retuning. We split the 10 audited blocks into two complementary 5-block calibration/evaluation halves, learn a single shell-calibrated threshold on one half, and apply it unchanged to the other half at nominal $\alpha = 10^{-3}$. Table 16 shows the result. Under that fixed threshold transfer, PD and Kasai collapse to zero held-out vessel-core recall, whereas the whitened matched-subspace specialist retains a small but nonzero held-out core TPR of 0.002 [0.000, 0.004] at a realized shell FPR of 0.001 [0.000, 0.002]. The effect is still modest, but it shows that the structural claim does not depend entirely on per-window threshold readjustment.

These real-IQ experiments are best interpreted as bounded structural and robustness audits rather than as definitive labeled *in vivo* task benchmarks. They should also not be read as a replacement for upstream motion correction or registration. Motion correction remains central in ULM and Doppler pipelines because residual motion and clutter leakage can dominate downstream vascular estimates [38, 40, 41]. The narrower claim here is that a local nuisance-aware detector head can make the exported score map less brittle to residual motion mismatch after ordinary residualization. That is still practically useful, because many acquisitions fail by producing unstable hot spots and threshold drift rather than by complete loss of flow signal.

Window	self-correlation base→matched-subspace	flow hit at bg tail 10^{-3} base→matched-subspace	clip fraction matched-subspace>base (correlation)	clip fraction matched-subspace>base (hit rate)
0:128	0.188 → 0.454	0.000 → 0.041	1.0	1.0
64:192	0.186 → 0.399	0.000 → 0.040	0.9	1.0
122:250	0.182 → 0.364	0.000 → 0.041	0.8	1.0

Table 17: Compact Shin rat-brain open real-IQ corroboration under controlled residual-motion stress (10 clips; 2 px amplitude; rigid registration enabled; optional shrink-only penalty disabled). "Self-correlation" is the Pearson correlation between each method's score map and its own no-motion reference after alignment and a boundary crop. "Flow-proxy hit rate" is evaluated at a matched background-proxy tail rate of 10^{-3} under a fixed, method-independent flow-proxy mask. The purpose of this table is not to claim task ROC on Shin, but to show that the same nuisance-robustness pattern appears on a second open real-IQ dataset.

Shin real-IQ audit. To avoid making the real-IQ story depend on a single open dataset, we also summarize the strongest existing Shin rat-brain motion-stress result from the supplement [35, 36]. Shin does not provide independent flow/background labels, so we do not treat it as a detection benchmark. Instead, under a prespecified low-frequency operating setting and a controlled residual-motion stress test, we ask whether the same qualitative robustness pattern seen on ULM recurs on a second open real-IQ source. It does: across all three audited windows, the matched-subspace score map is more stable than the MC–SVD baseline map under the same 2 px residual-motion stress, and the matched background-tail operating point yields a nonzero flow-proxy hit rate only for the matched-subspace detector.

Taken together, the real-data results are consistent but intentionally limited in scope: the Gammex phantom provides structurally labeled phantom evidence, the ULM motion audit provides open real-IQ robustness evidence, the ULM structural audit adds a same-acquisition structural check using a localization-derived vascular reference, and the Shin real-IQ audit provides a second open real-IQ corroboration under controlled residual-motion stress. A compact same-residual score audit across those real-IQ bundles is kept in the Supplementary Material so the main paper can stay focused on the structurally labeled and deployment-relevant endpoints. Additional label-free audits, such as whole-brain PD-only retrospectives, Shin contract summaries, and extended motion variants, are likewise provided in Supplementary Material.

6 Discussion

What the detector family adds

The main message of this study is simple: post-beamformed fUS benefits from a better detector, not only from better residualization. Ordinary clutter suppression can still leave a small number of structured false detections that dominate the right tail of the score distribution and shape map trustworthiness. The clearest general result in the paper is that localized band-limited matched-subspace scoring is the best-performing fixed head among the evaluated downstream heads on the held-out structural benchmark. The labeled brain simulations and the structurally labeled phantom explain why: score construction itself matters in the strict tail, and nuisance-aware local scoring can preserve measurable sensitivity where several conventional score heads collapse. Just

as importantly, the same-residual ablations show what the contribution is not. The main labeled-brain gain is not reproduced by total whitened power without Doppler band partition, so the paper should be read as a detector-head design result, not as a generic endorsement of whitening or score engineering.

Those same ablations also define the current boundary of the method family. On the labeled brain structural stress tests, most of the gain comes from band-limited matched-subspace scoring rather than from whitening alone. Whitening therefore helps in some regimes but not in all of them. It is not the main source of gain in the labeled brain structural stress tests, but it is useful in the artifact-heavy phantom, in the ULM structural audit, and in selected functional and stress-frontier settings. On the held-out structural benchmark, the unwhitened head remains the strongest fixed nuisance-control choice. The adaptive modes still matter, but in a narrower way: the adaptive switch appears to preserve fixed-head behavior in cleaner structural settings while recovering some whitening benefits in selected clutter-heavy conditions, and the fully whitened mode is most convincing where the extra covariance adaptation clearly earns its complexity. The practical result is a detector family with a clear fixed default and two more specialized extensions, not a claim that one whitened score is best everywhere.

One deployment point is worth stating explicitly. The strict-tail operating points in the labeled experiments are calibrated using known negative sets, but a scanner does not have frame-by-frame labeled negatives at runtime. In this paper, that tail calibration is an *evaluation protocol*, not the intended clinical operating rule. The more deployment-relevant evidence now appears in the main text: a separate-bank threshold transfer on the held-out SIMUS/PyMUST benchmark and a complementary block-split fixed-calibration transfer on the ULM structural audit, both without per-window retuning on the evaluation data. Those checks do not solve cross-session or cross-scanner thresholding outright, but they do show that the central nuisance-control claim survives a genuinely fixed-threshold test on both the main benchmark and one real-IQ endpoint. The remaining external step is prospective validation of a practical thresholding workflow on raw *in vivo* IQ without retrospective labels.

Real-data scope and clinical interpretation

The optional shrink-only regularizer is deliberately conservative. It can only shrink scores, it preserves an explicit do-not-penalize set, and it is designed to remain inert when the side information is weak or ambiguous. The calculi audit is therefore meant to show bounded behavior rather than dramatic uplift: the regularizer modestly compresses the extreme tail while leaving protected locations unchanged. That is the right role for this layer in a medical signal-processing pipeline, where bounded failure modes matter more than aggressive apparent gains. It is also why we do not make it part of the main headline claim.

A broader practical limitation is data availability. Most public *in vivo* fUS releases do not preserve the raw high-PRF complex IQ and metadata needed for independent detector evaluation after beamforming. For that reason, the strongest quantitative efficacy evidence in this paper still comes from labeled simulations and structurally labeled phantom data, while the open real-IQ sections are framed more narrowly as robustness, same-residual, and map-stability audits. The ULM structural audit narrows that gap. On a vessel-core versus perivascular-shell endpoint derived from a published localization rendering, the fully whitened matched-subspace specialist is modestly but consistently stronger than PD and Kasai, and the 64-frame no-registration implementation remains below the acquisition-time budget on an RTX 4080 SUPER. The effect size is modest, and we mean

that literally. What matters is that the direction is consistent on a same-acquisition structural audit using a localization-derived vascular reference, with labels derived from a different physical process than the Doppler detector itself. That is still not independent multimodal validation, and it is not a claim about functional activation or hemodynamic-response recovery, but it is materially stronger than a purely label-free map-stability result.

The important field-level point is that this remaining gap is no longer just a gap in this manuscript. Public resources are now beginning to supply the right ingredients in separate places: simultaneous fUS–Neuropixels recordings in awake mice, raw fUS plus two-photon multimodal datasets, and translational human task-fUS studies with cranial windows or sonolucent implants [8,22–25]. What is still missing is a standardized open benchmark that packages those ingredients in a detector-ready way for post-beamformed evaluation across repeated sessions and fixed-threshold transfer.

We therefore interpret the present real-data results as evidence for improved vascular/nuisance discrimination at the detector stage rather than as direct evidence of improved biological inference. Studies linking fUS to neural activity already show that this translation depends on neurovascular coupling and temporal transfer structure rather than on detector quality alone [26]. Even with that remaining limitation, the clinical interpretation is already useful: this is a step toward more trustworthy fUS maps in realistic nuisance and mild-motion conditions, because it reduces artifactual hot spots and nuisance leakage without requiring a new acquisition pipeline. The intended gain is not merely better vessel conspicuity, but more stable decision-relevant maps at matched parenchymal sensitivity when rare false positives would otherwise dominate interpretation.

Detector policy and next validation step

The current evidence supports a simple three-part policy. The fixed band-limited matched-subspace detector is the main result and default for held-out structural use. The adaptive mode is the practical extension when one wants additional clutter suppression without forcing whitening everywhere. The fully whitened variant remains a bounded specialist mode for the regimes in which full covariance adaptation already earns its complexity, especially the across-view phantom, the held-out functional audit, and the bounded ULM structural audit. The main-text held-out stress frontier is the clearest reason this hierarchy is needed at all: under the cardiac-reference and shortest-ensemble rows, the fixed matched-subspace default is already best, whereas the harsher pulsation rows and the intermediate short-ensemble row are won by the fully whitened specialist. The ULM structural audit shows the same principle on real data for a different reason: vessel-core concentration relative to the immediately adjacent shell favors the fully whitened matched-subspace specialist, whereas nuisance-control on the held-out SIMUS/PyMUST benchmark favors the simpler fixed head. A compact policy table is kept in the Supplementary Material so the main paper can remain focused on the core evidence rather than on deployment bookkeeping.

7 Conclusion and future work

We presented a localized matched-subspace detector framework for post-beamformed fUS and ultrafast Doppler data, centered on a band-limited fixed head and extended by an adaptive whitening switch plus a fully whitened specialist mode. Across the evaluated labeled brain stress tests and the Gammex structural-label phantom, this framework retained nonzero sensitivity at strict low-

FPR operating points where PD/Kasai-based comparators could collapse. Simple same-residual ablations also showed that this gain is not explained by whitening alone: total whitened power without Doppler band partition still collapses in the labeled brain strict tail, whereas the fixed band-limited ratio remains strong. On the held-out SIMUS/PyMUST benchmark, the simpler band-limited matched-subspace detector was the best-performing fixed head among the evaluated downstream heads. A held-out SIMUS stress confirmation then showed that under cardiac-like pulsation and short-ensemble pressure, the adaptive and fully whitened extensions add value on the rows where nuisance pressure is highest. A separate fixed-calibration transfer audit on that same benchmark showed that thresholds learned on separate calibration seeds still preserve the fixed head's nuisance-control advantage on held-out evaluation seeds without per-window retuning. On real in-vivo rat-brain IQ, a same-acquisition structural audit using a localization-derived vascular reference from the published PALA rendering showed a modest structural advantage for the fully whitened matched-subspace specialist over PD and Kasai on a vessel-core versus perivascular-shell endpoint; the same 64-frame no-registration configuration also remained below the acquisition-time budget on an RTX 4080 SUPER, and a complementary 5-block fixed-calibration transfer retained a small but nonzero held-out core-detection advantage at near-target shell FPR. This in-vivo claim is about structural vascular fidelity rather than functional activation or hemodynamic-response recovery. The real-IQ evidence therefore now includes a quantitative structural audit on real in-vivo data, but it remains limited rather than definitive independent-task validation.

The practical implication is therefore clear. Nuisance-aware matched-subspace scoring is the main deployed consequence of this framework; the adaptive mode is the clearest compromise when additional clutter suppression is needed, and the fully whitened mode is retained only for the settings in which it already earns its complexity. The near-term clinical implication is pragmatic: a nuisance-robust local detector may help move fUS and ultrafast Doppler outputs from "informative under careful conditions" toward "reliably interpretable under messier real conditions," especially when rare false positives dominate downstream decisions. The main next external test is prospective labeled *in vivo* validation on raw high-PRF IQ with independent anatomical or task anchors and a practical non-retrospective thresholding workflow.

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. All code used to generate the reported tables and figures is versioned; the exact commit hash and release information are recorded in the reproducibility manifest (Supplementary Material). A minimal CUDA-enabled software environment and container recipe are provided in the repository. End-to-end regeneration commands for the reported figures and tables are collected in the repository and summarized in the reproducibility appendix.

Code repository. The reference implementation and reproduction scripts are available at `https://github.com/ArthurShune/stap-for-fus`. The canonical commit hash and release information are recorded in the reproducibility manifest (Supplementary Material).

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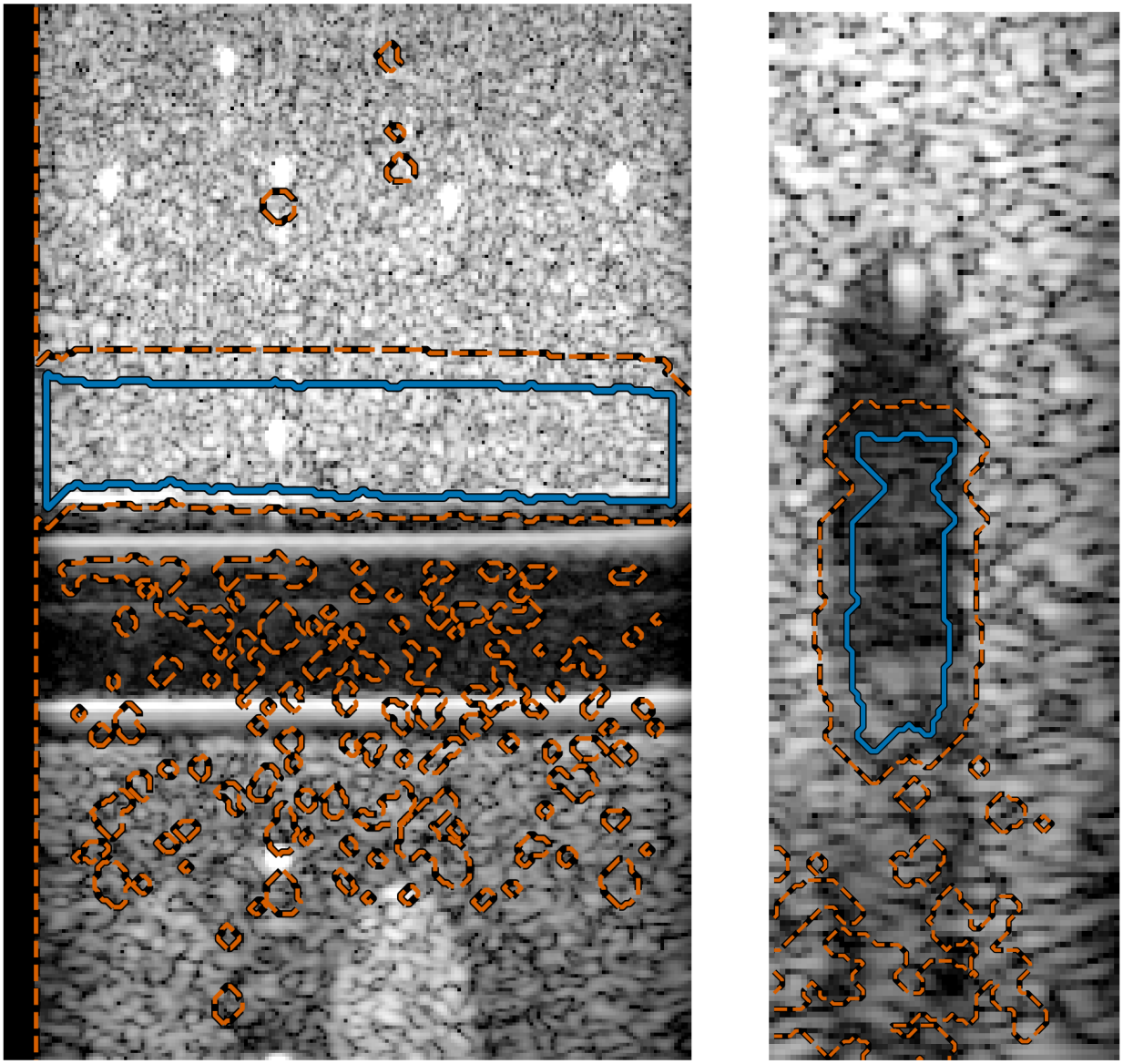

Figure 9: Gammex flow phantom structural evaluation masks derived from B-mode only. Left: along-linear view; right: across-linear view. The lumen mask is intentionally eroded to avoid wall/partial-volume pixels, and the background mask excludes a guard band around the lumen and bright reflectors. Images show the CFM-aligned B-mode ROI with contour overlays of the tube lumen (flow) mask (blue) and background mask (orange, dashed) used for structural-label ROC reporting. Reproducibility details are provided in Supplementary Material.

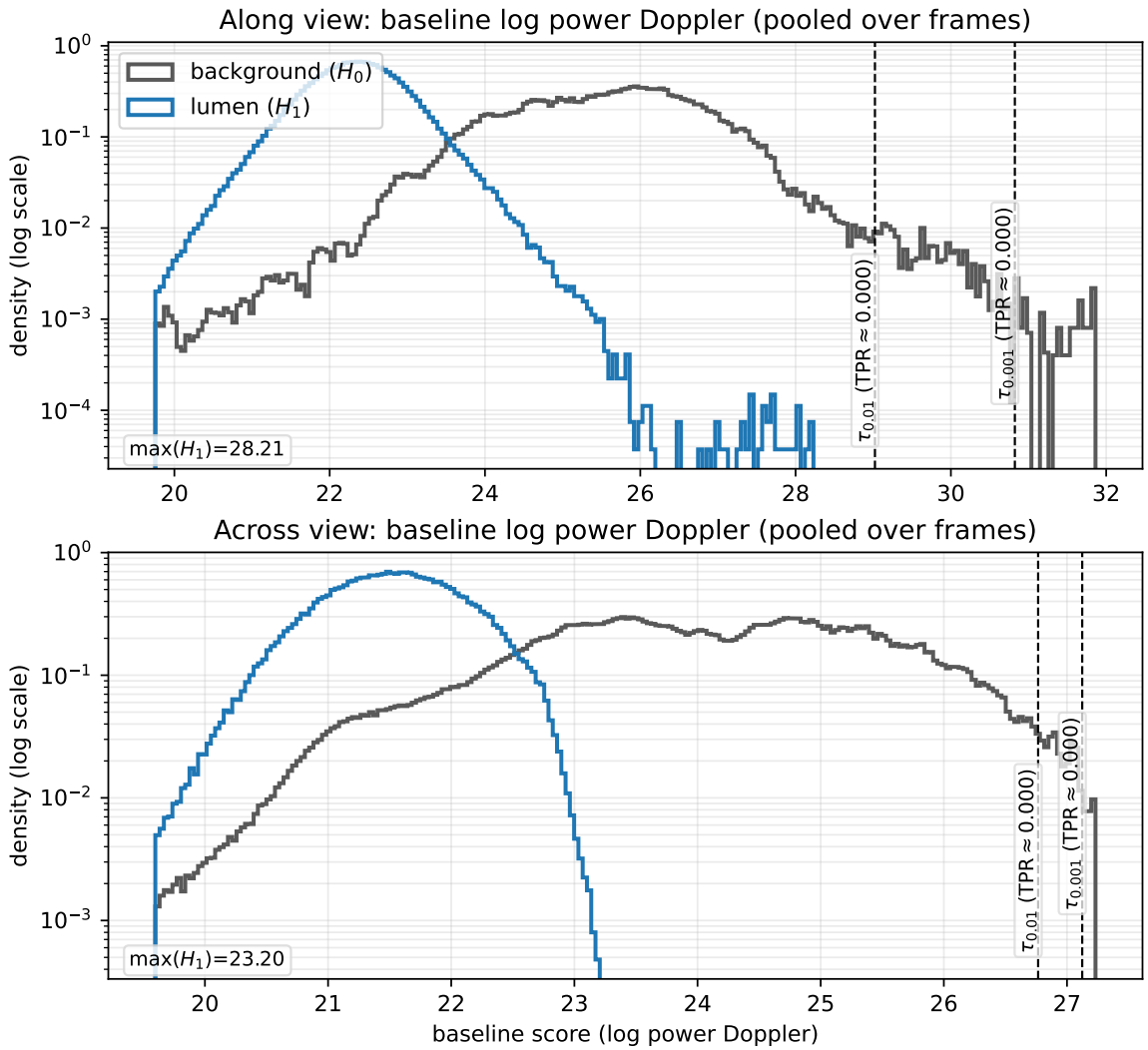

Figure 10: Why conventional baselines can have zero TPR at ultra-low FPR on Gammex short-ensemble data. The plot shows pooled score distributions for the baseline log power Doppler comparator on lumen (H_1; structural tube mask) versus background (H_0; structural background mask), for both along and across views. Vertical lines indicate background-calibrated right-tail thresholds τ_α at $\alpha \in \{10^{-2}, 10^{-3}\}$ (the main text reports $\alpha \leq 10^{-3}$; $\alpha = 10^{-2}$ is shown here for context). On both views the background tail threshold lies above $\max(H_1)$, so $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$ is numerically zero at these tail operating points.

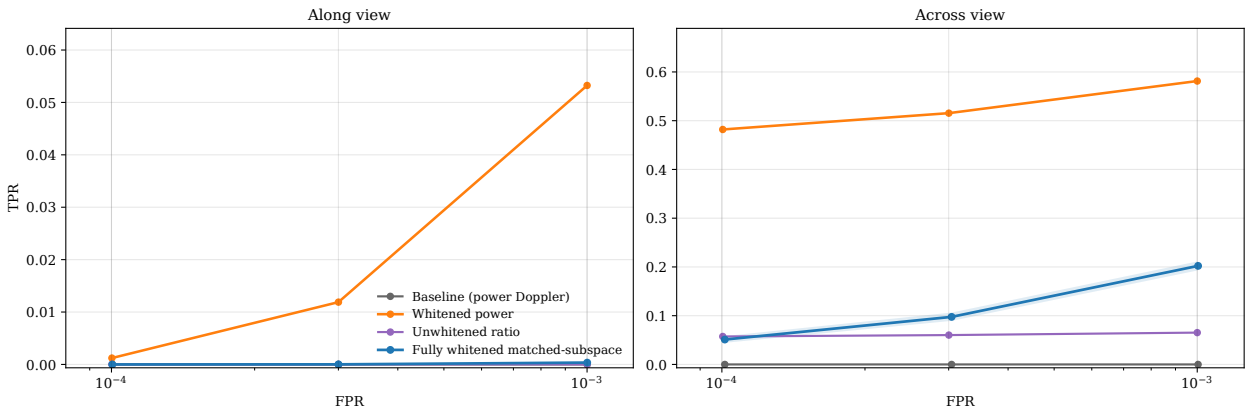

Figure 11: Gammex flow phantom same-residual score comparison in the strict low-FPR region used for the main phantom table. Curves connect the three reported operating points for baseline power Doppler, whitened power, the fixed matched-subspace detector, and the fully whitened matched-subspace detector before the optional shrink-only penalty. The two panels use panel-specific TPR ranges so that the operating-point differences are readable.

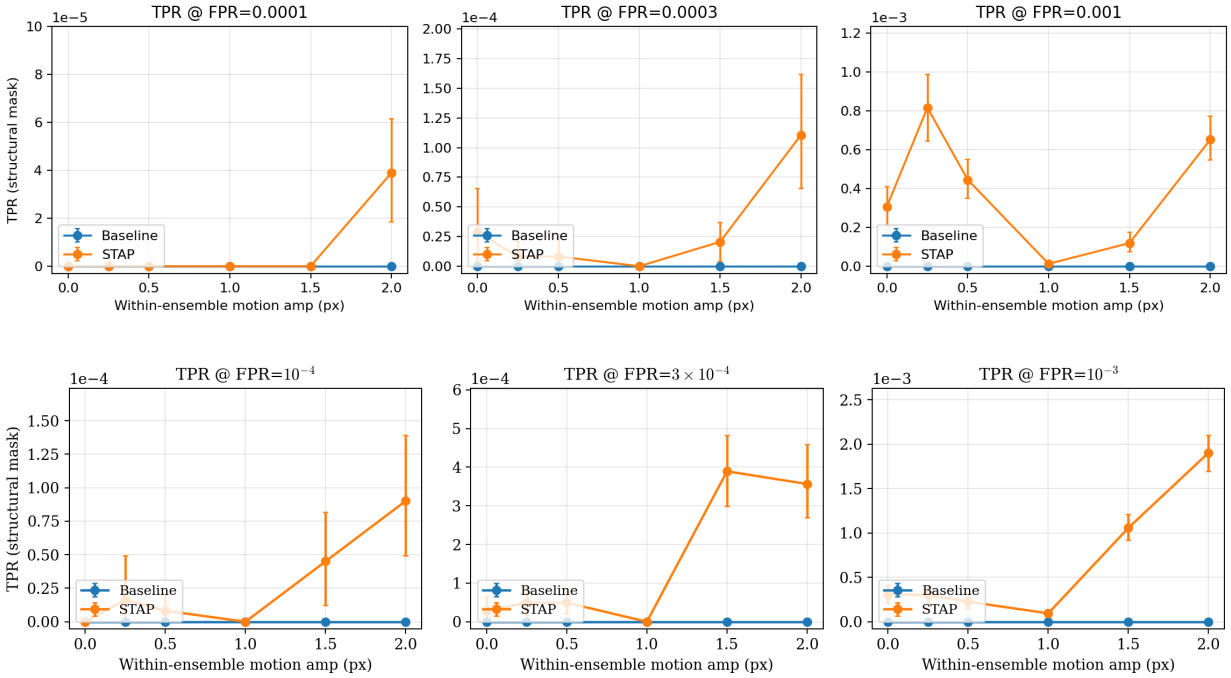

Figure 12: Gammex flow phantom within-ensemble motion ladder on the along-linear PRF=2500 sequence (85 frames). Left: random-walk per-shot translation; right: step translation. Each sub-panel reports TPR vs motion amplitude at a fixed FPR target using B-mode structural masks, with the y-axis zoomed to the observed TPR range so that small changes remain visible. Shaded bands are 25–75% bootstrap intervals over cine frames. Reproducibility details are provided in Supplementary Material.

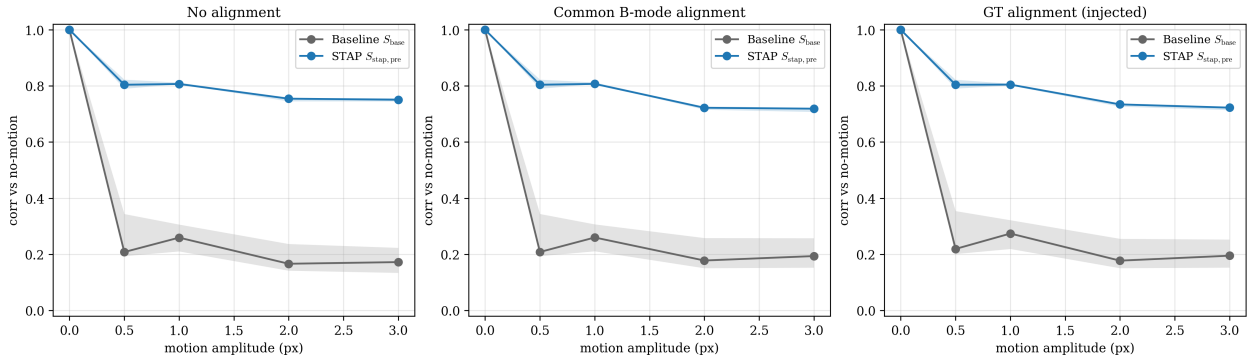

Figure 13: ULM 7883227 motion sweep (blocks 1–3; frames 0:128; frozen baseline $e = 0.975$): `brainlike` motion (rigid + elastic). Each panel reports correlation vs the no-motion reference under a different alignment mode (none / common B-mode / injected GT proxy) for the baseline score S_{base} and the matched-subspace score $S_{\text{stap,pre}}$; shaded bands show median±IQR over blocks. Reproducibility details are provided in Supplementary Material.